

ChipStain: uncertainty-aware label-free nuclear staining, towards organ-on-a-chip imaging

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Code: <https://github.com/hamidhosen42/AI4S-Open-Innovation-AI-for-Life-Science> · **Report version:** 2026-10-07

Abstract. Nuclear read-outs on an organ-on-a-chip (cell number, proliferation, viability) start from a nuclear stain, but fixation ends the experiment, live DNA dyes add phototoxic exposure, and reporter lines are impractical for primary or iPSC-derived tissue. In-silico labeling (ISL) predicts the stain from bright-field, but a point prediction gives no warning when it is wrong. ChipStain predicts the H2B nuclear channel from one bright-field image together with a per-pixel uncertainty σ (U-Net with an ImageNet ResNet-34 encoder, β -NLL heteroscedastic head, $8\times$ test-time augmentation). We validate the uncertainty, not only the image, on a public HeLa time-lapse in a 96-well plate (held-out well, 125 images, 3 training seeds; no chip data were available to us). Against a matched control — the same ImageNet U-Net trained with L1 and the same augmentation — ChipStain has the same correlation with the real stain (r 0.768 vs 0.768) at 11 % higher pixel error, and its σ ranks pixel errors better (Spearman ρ 0.47 vs 0.32; AUSE 0.177 vs 0.248; in every seed). σ is approximately calibrated without recalibration (nominal 95 % intervals cover 92 % of pixels). Under simulated blur or a phase-contrast input every model fails and σ is not a reliable drift alarm, but a cheap input check (image focus plus encoder-feature distance) flags all of these images, at 16 % false alarms on clean images. Gating frames by σ lowers the mean bias of label-free doubling times from 9.1 % to 4.1 %, mostly by routing failing fields to review. For counting alone, a direct bright-field segmenter is more accurate (Cellpose F1 0.831 vs 0.708); ChipStain's value is a fluorescence-like image with a calibrated trust signal. Open code and weights; CPU inference.

1. Problem definition and application scenario

1.1 Why a label-free nuclear channel matters on a chip

Organ-on-a-chip (OoC) systems rebuild tissue-level structure and function in microfluidic devices and are imaged repeatedly — often for days or weeks — by transmitted-light microscopy. Nearly every quantitative read-out derived from those images starts by locating nuclei: cell number, confluence, proliferation, apoptosis and pyknosis, nuclear morphology, compound dose-response. That step is usually done on a fluorescent DNA stain or a fluorescent histone reporter, each with a cost in the OoC setting:

- **Fixed stains end the experiment.** DAPI after fixation gives one time-point per chip; a time course needs one chip per time-point.
- **Live DNA dyes perturb the cells they measure.** Repeated excitation is phototoxic [1]; Hoechst 33342 is phototoxic in time-lapse imaging [11], and the live-cell dye SiR-Hoechst induces DNA-damage responses and impairs cell-cycle progression [12].
- **Reporter lines must be engineered** — routine for a cell line such as the HeLa H2B line used here, but often impractical for primary or iPSC-derived tissue, e.g. the neural cultures that are the focus of the challenge's supporting organisation.
- **Each fluorescence channel spent on nuclei** is unavailable for a functional reporter (e.g. a Ca^{2+} indicator), and every perfused reagent adds handling steps and bubble/contamination risk in small channels.

In-silico labeling (ISL) — predicting the fluorescence image from the label-free image with a neural network [2, 3] — can reduce routine staining once a paired training set exists for the system (§8), and is one of the three impact areas named by the organisers. What keeps ISL out of quantitative decisions is **trust**: a network returns an image for every input, including inputs unlike its training data, and nothing in a point prediction tells the biologist which pixels, nuclei or frames are wrong. ChipStain therefore asks a narrower and more useful question than "can we predict the stain?": **can we predict it and say — per pixel, per nucleus and per frame — how much to trust it, and does that trust signal hold up against simple alternatives?**

Target users are OoC and cell-culture laboratories and CROs running live or high-content assays, and platform developers who need a quality gate before automated analysis. **Scope and honesty statement.** All quantitative validation uses public data: 2-D HeLa monolayers in a 96-

well imaging plate (main results, 60 h time-lapse) and human iPSC-derived motor-neuron cultures in a dish (transfer test). **No organ-on-a-chip device data were available to us**; OoC is the target deployment, and §6.7 tests how the model behaves under the imaging shifts a chip introduces.

Value case (illustrative arithmetic, stated assumptions)

A neural OoC culture imaged every 30 min for 14 days at 4 fields per chip needs $14 \times 48 \times 4 = 2,688$ **nuclear-channel exposures per chip**; a label-free nuclear channel removes all of them and frees that channel for a functional reporter. Fixed DAPI would instead need one chip per time-point. Neurons are post-mitotic, so label-free nuclear counts and nuclear condensation over weeks serve as a viability/neurotoxicity end-point without repeated dye exposure. The cost of being wrong is mitigated by a frame-level σ gate (§6.10), which routes frames with unusually high mean σ to a person instead of a growth curve.

2. Related work and contribution

In-silico labeling. Christiansen et al. [2] predicted multiple fluorescent labels from transmitted-light z-stacks, outputting for every pixel a *discrete probability distribution over 256 intensity values*. Ounkomol et al. [3] (“fnet”) trained a deterministic 3-D U-Net with a mean-squared-error loss; adversarial virtual staining followed for histology [13]. **Uncertainty in bioimage restoration.** CARE [14] predicted per-pixel Laplace parameters and used the disagreement of an ensemble to flag unreliable regions; a recent diffusion-based ISL method derives hallucination maps from the variance across denoising steps [15]. **Uncertainty estimation.** Heteroscedastic regression models aleatoric noise [4]; its maximum-likelihood training is unstable and β -NLL [8] fixes this; deep ensembles [16] and test-time augmentation [17] add model-related uncertainty; sparsification curves and the area under the sparsification error (AUSE) [18] score how well an uncertainty ranks errors.

Contribution. ChipStain introduces no new uncertainty mechanism. Its contributions are: (i) a β -NLL-stabilised heteroscedastic ISL model with test-time augmentation for bright-field $\rightarrow$ nuclei, released as an open, CPU-capable tool; and (ii) a validation of its uncertainty that the ISL literature rarely reports — against an equally strong TTA baseline, against uncertainty-free proxies (predicted intensity, edge strength, cell density), at pixel, nucleus and frame level, under imaging shift, on a biological read-out (doubling time) with a gate fixed on validation data, and on neural cultures. We report where σ helps and, equally, where it does not.

3. Data

3.1 Main dataset: source, licence, usage note

Automatic labelling of HeLa "Kyoto" cells using Deep Learning tools by Romain Guet (EPFL Bioimaging & Optics Platform), Zenodo 10.5281/zenodo.6140064, CC BY 4.0 (<https://creativecommons.org/licenses/by/4.0/>) [5], derived from the time-lapse *HeLa "Kyoto" cells under the scope* (10.5281/zenodo.6139958, CC BY 4.0) [6]. HeLa "Kyoto" cells stably expressing EGFP- α -tubulin and mCherry-H2B were grown as a 2-D monolayer in a CellCarrier Ultra 96-well plate (PerkinElmer) and imaged on a PerkinElmer Operetta (20 $\times$ /NA 0.8) every 15 min for 60 h. The archive provides, for 5 time-points (frames 1, 10, 50, 100, 150 $\approx$ 0–37 h) of every field, a 540×540 px 16-bit bright-field image, two digital-phase-contrast renderings, a two-channel fluorescence image and StarDist nuclei labels computed automatically on the H2B channel [10]. Input is bright-field; target is mCherry-H2B (channel 1: correlation with the binarised nuclei labels 0.787 vs 0.416 for tubulin over the 200 training images; `scripts/check_channels.py`). **Usage note.** The provider asks that the fluorescence not be reused *with the derived labels* for training and notes that the automatic labels are imperfect. We comply: we train only bright-field $\rightarrow$ fluorescence, never on the labels, and use the StarDist labels solely as reference objects for evaluation. Segmentation F1 and count errors therefore measure agreement with StarDist run on the real stain, not with manual ground truth.

3.2 Splits without leakage

Split	Wells	Fields	Images	Use
Train	R05-C05, R05-C07	0–19 ($\times 5$ time-points)	200	optimisation
Validation	R05-C05, R05-C07	20–24 ($\times 5$ time-points)	50	checkpoint selection; gate and recalibration thresholds
Test	R05-C03 (held-out well, same plate and imaging session)	all 25 ($\times 5$ time-points)	125	all reported numbers

All time-points of a field stay on the same side of every split. All three wells come from one plate and one imaging session (file prefix 20210904_TL2), so the hold-out measures well-to-well generalisation, not generalisation across experiments, days or microscopes. Because the 125 test images are 25 fields $\times$ 5 time-points of one time-lapse, the field ($n = 25$) — not the image — is the unit of statistical testing (§6.2).

3.3 Additional evaluation data

60 h time-lapse. The full 240-frame recording of test field R05-C03-F0 (10.5281/zenodo.6139958, CC BY 4.0) is used only for evaluation (§6.10); five of its frames are also test images. **Neural cultures.** Condition A of the in-silico-labeling data [2] (human iPSC-derived motor neurons; bright-field z-stacks and widefield Hoechst nuclear stain (channel named DAPI_WIDEFIELD in the release); 22 training and 3 test wells; CC BY 4.0) is used for the transfer test (§6.11). **Direct-segmentation competitor.** The Cellpose models published by the dataset author (10.5281/zenodo.6140111, CC BY 4.0), trained to segment nuclei directly from bright-field (§6.8).

3.4 Pre-processing, compliance and ethics

Bright-field images are normalised per image with their 1st/99th percentiles ($\approx [-2, 2]$), making the model insensitive to camera offset and gain; the target is scaled with fixed constants from the training H2B distribution, $(I - 600)/(20\,000 - 600)$, clipped to $[0, 1]$, and all image metrics are computed in this space. Training uses random 256×256 crops with 90° rotations and flips; evaluation uses full 540×540 images. Figures show percentile-normalised, colour-mapped renderings; every image is listed with its source and modifications in Appendix E. All data are public and CC BY 4.0. HeLa is an established human-derived cell line (taken from Henrietta Lacks in 1951 without her consent); the images contain no genetic, clinical or personal information. No clinical, personal or restricted data are used.

4. Method

4.1 Backbone

A U-Net [7] built with segmentation_models_pytorch [9]; ResNet-34 encoder [19] initialised from ImageNet [20]; five-stage decoder with skip connections; one input channel; ≈ 24.4 M parameters.

4.2 Heteroscedastic output and β -NLL

The last layer outputs two maps: the predicted mean $\mu(x)$ and log-variance $s(x) = \log \sigma^2(x)$ of a per-pixel Gaussian over the target y . The Gaussian negative log-likelihood

$$L_{NLL} = \frac{1}{2} \sum_p [s_p + (y_p - \mu_p)^2 \cdot e^{-s_p}]$$

is known to be unstable: pixels that are hard early in training receive a large σ , the gradient on their mean is down-weighted by e^{-s} , and they are under-fitted [8]. β -NLL [8] multiplies each pixel's loss by the *detached* factor $(\sigma^2)^\beta$; we use $\beta = 0.5$ ($\beta = 0$ recovers the NLL; at $\beta = 1$ the gradient with respect to μ equals the MSE gradient while σ^2 is still fitted). Because the loss is Gaussian, the mean head is trained with a σ -weighted squared error rather than the L1 loss of our baseline — a confound we test explicitly (§6.3). s is clamped to $[-10, 4]$. σ is the model's learned (aleatoric) uncertainty.

4.3 Test-time augmentation, ensembles and the reported σ

At inference the network is applied to the eight dihedral transforms of the input (4 rotations $\times$ optional flip); outputs are mapped back and $\bar{\mu}$ is their mean. The reported uncertainty is $\sigma_{\text{tot}} = \sqrt{(\text{mean of the eight } \sigma^2 + \text{variance of the eight } \mu)}$. We call the second term **TTA disagreement**: it measures how much the prediction changes under symmetries the model should respect, a partial proxy for model uncertainty, not a Bayesian epistemic estimate [17]. For the plain U-Net baseline, the TTA disagreement alone is its uncertainty. We also evaluate **3-seed deep ensembles** [16] built from the cached predictions of the three training seeds ($\sigma^2 = \text{variance of member means} + \text{mean member } \sigma^2$).

4.4 From pixels to decisions

Per nucleus: nuclei are segmented on the prediction (Gaussian blur, Otsu threshold, distance-transform watershed); each nucleus receives the mean σ inside it. **Per frame:** the mean σ of the image, divided by the median over frames of the same time-point (cell density grows with time, and σ with density); a frame is flagged when this ratio exceeds its 95th percentile **on the validation split** — a 5 % false-alarm budget fixed without test labels.

4.5 Evaluation protocol

Fidelity: MAE, PSNR, SSIM, Pearson r against the real H2B. **Downstream:** object-level F1 at IoU ≥ 0.5 against the StarDist reference nuclei; the same pipeline run on the real H2B gives the reference level a perfect prediction would reach. **Uncertainty ranking:** Spearman $\rho(\sigma, |\text{error}|)$ on a fixed 200 000-pixel sample per image; AUSE [18] (area between the sparsification curve and the oracle curve, normalised; lower is better); and the MAE drop after removing the 20 % highest- σ pixels — each also computed for two uncertainty-free proxies taken from the prediction itself: predicted intensity μ and its edge strength $|\nabla\mu|$. **Calibration:** empirical coverage of $\mu \pm \sigma$. **Statistics:** 3 training seeds per configuration. Inference rests on 95 % confidence intervals from a hierarchical bootstrap that resamples seeds and fields, and on per-seed field-level Wilcoxon tests; a difference is called robust only if its CI excludes zero, no seed goes the other way and at least two seeds are significant on their own. A pooled field-level Wilcoxon (Holm-corrected) is reported for the uncertainty metrics only.

5. Implementation

Item	Value
Software	Python 3.12.11, PyTorch 2.14.0, segmentation_models_pytorch 0.5.0, scikit-image 0.26.0, SciPy 1.18.1, NumPy 2.5.2 (exact pins: requirements-lock.txt)
Optimiser	AdamW (weight decay 1e-4), one-cycle LR (peak 1e-3 from scratch, 5e-4 with the pretrained encoder; 10 % warm-up), gradient clipping 1.0
Training	40 epochs $\times$ 2 random-crop passes over the 200 training images; batches of 8×256^2 crops; checkpoint = best validation Pearson + SSIM
Seeds	0, 1, 2 for every configuration (main and ablation arms)
Hardware / time	Apple M5 laptop (Metal/MPS): 9–28 min per 40-epoch run (measured from the run logs)
Inference	runs on a CPU (CUDA/MPS used if present): ≤ 0.5 s per 540×540 image single pass, ≤ 2.4 s with 8 $\times$ TTA (measured on Apple M5 CPU, 8 threads)
External downloads	datasets from Zenodo / Google Cloud (public); ResNet-34 ImageNet weights from the Hugging Face Hub (training only); released checkpoint from the GitHub release. No paid service, API or proprietary hardware.
Entry points	scripts/train.py, scripts/evaluate.py, scripts/inference.py, demo/app.py; analyses in scripts/ (§9)

6. Experiments and results

6.1 Main results over three training seeds

Model	Pearson r ↑	SSIM ↑	MAE ↓	PSNR ↑	seg-F1 ↑	$\rho(\sigma, \text{err})$ ↑	AUSE ↓	MAE drop, top-20 % σ removed ↑
Scratch U-Net (L1, lr 1e-3)	0.749 ± 0.027	0.826 ± 0.003	0.0299 ± 0.0006	24.15 ± 0.24	0.648 ± 0.089	—	—	—
ImageNet-L1 U-Net (L1, lr 5e-4)	0.755 ± 0.005	0.823 ± 0.005	0.0296 ± 0.0003	24.25 ± 0.07	0.669 ± 0.021	—	—	—
Scratch U-Net + TTA (σ = view s.d.)	0.764 ± 0.024	0.831 ± 0.003	0.0294 ± 0.0006	24.36 ± 0.25	0.665 ± 0.083	0.283 ± 0.071	0.258 ± 0.025	40.5 ± 3.2 %
ImageNet-L1 U-Net + TTA (σ = view s.d.) — matched control	0.768 ± 0.006	0.827 ± 0.005	0.0291 ± 0.0003	24.47 ± 0.10	0.690 ± 0.019	0.317 ± 0.037	0.248 ± 0.011	42.9 ± 1.8 %
ChipStain (β -NLL, β = 0.5)	0.757 ± 0.015	0.810 ± 0.018	0.0324 ± 0.0013	23.92 ± 0.36	0.697 ± 0.012	0.430 ± 0.006	0.303 ± 0.072	45.6 ± 1.6 %
ChipStain + TTA (full)	0.768 ± 0.012	0.814 ± 0.022	0.0323 ± 0.0007	24.11 ± 0.32	0.708 ± 0.011	0.469 ± 0.027	0.177 ± 0.021	49.7 ± 4.2 %
Real fluorescence, same segmentation pipeline (reference level, not a bound)	—	—	—	—	0.765	—	—	—

Mean $\pm$ s.d. over 3 training seeds; each seed is the mean over the 125 test images of well R05-C03. The released checkpoint is seed 0 (ChipStain + TTA: Pearson r 0.779, SSIM 0.808, seg-F1 0.718, $\rho(\sigma, \text{err})$ 0.441). The real-fluorescence row runs the same watershed on the real H2B image; it is a reference level, not a strict bound.

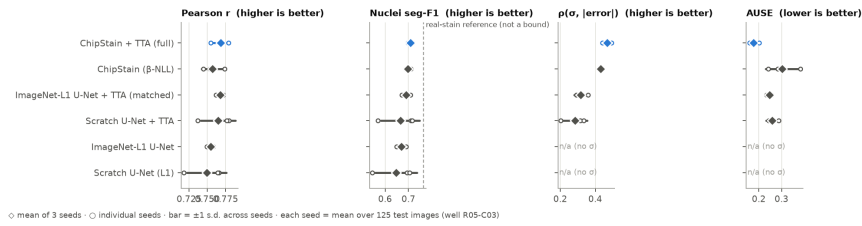


Figure 1. The six main configurations over three training seeds: open circles are seeds, diamonds their mean, bars ± 1 s.d.; the dashed line is the real-stain reference level (not a bound). ChipStain + TTA (blue) is the full method.

6.2 Like-for-like comparison and statistics

The like-for-like control for the full method is an **ImageNet-encoder L1 U-Net with the same 8× TTA and the same learning rate** (its uncertainty is the TTA disagreement): it differs from ChipStain only in the probabilistic head and loss. Differences are ChipStain + TTA minus that control, with a 95 % CI from a bootstrap that resamples both seeds and fields, and per-seed field-level tests.

- **Correlation with the real stain:** Pearson r +0.000 (95 % CI -0.019 to +0.013; significantly better in 2/3 seeds, worse in 0/3) — identical within run-to-run variation.
- **Pixel fidelity:** MAE +0.0032 (95 % CI +0.0012 to +0.0056; significantly better in 0/3 seeds, worse in 2/3) (+11 %); SSIM -0.013 (95 % CI -0.032 to +0.010; significantly better in 0/3 seeds, worse in 2/3). The MAE cost is small but consistent (its CI excludes zero); §6.3 traces it to the change of loss family (L1 $\rightarrow$ squared error), not to the variance head.
- **Downstream nuclei F1:** +0.018 (95 % CI -0.009 to +0.042; significantly better in 2/3 seeds, worse in 0/3).
- **Uncertainty ranking:** $\rho(\sigma, |\text{error}|)$ +0.152 (95 % CI +0.106 to +0.201; significantly better in 3/3 seeds, worse in 0/3; better in 25/25 fields; pooled Holm $p < 0.001$); AUSE -0.070 (95 % CI -0.095 to -0.045; significantly better in 3/3 seeds, worse in 0/3; better in 25/25 fields; pooled Holm $p < 0.001$). This is the clearest and most consistent result of the study.

Against the scratch U-Net + TTA (from-scratch encoder, learning rate $1e-3$ — a weaker and less matched baseline) the uncertainty advantage is similar: ρ +0.186 (95 % CI +0.120 to +0.264; significantly better in 3/3 seeds, worse in 0/3; better in 25/25 fields; pooled Holm $p < 0.001$); AUSE -0.081 (95 % CI -0.114 to -0.049; significantly better in 3/3 seeds, worse in 0/3; better in 25/25 fields; pooled Holm $p < 0.001$).

Without TTA, ChipStain's learned σ alone still ranks errors better than the control's TTA disagreement in ρ , but **not** in AUSE (0.303 vs 0.248; no robust difference): the learned head and the TTA term are complementary, and the full method needs both.

A vs B	metric	A	B	A - B (95 % CI, seeds + fields)	seeds sig. better / worse	p (pooled, Holm; σ metrics)
ChipStain + TTA vs Scratch U-Net + TTA	Pearson r	0.7681	0.7644	+0.0037 (-0.0211, +0.0304)	1 / 0 of 3	—
ChipStain + TTA vs Scratch U-Net + TTA	SSIM	0.8137	0.8310	-0.0173 (-0.0361, +0.0061)	0 / 2 of 3	—
ChipStain + TTA vs Scratch U-Net + TTA	MAE	0.0323	0.0294	+0.0029 (+0.0009, +0.0054)	0 / 2 of 3	—
ChipStain + TTA vs Scratch U-Net + TTA	psnr	24.1148	24.3614	-0.2466 (-0.7972, +0.1756)	0 / 0 of 3	—
ChipStain + TTA vs Scratch U-Net + TTA	nuclei F1	0.7082	0.6651	+0.0431 (-0.0171, +0.1351)	1 / 0 of 3	—
ChipStain + TTA vs Scratch U-Net + TTA	$\rho(\sigma, \text{error})$	0.4689	0.2832	+0.1857 (+0.1201, +0.2641)	3 / 0 of 3	4.4e-06
ChipStain + TTA vs Scratch U-Net + TTA	AUSE	0.1775	0.2582	-0.0807 (-0.1136, -0.0493)	3 / 0 of 3	4.4e-06
ChipStain + TTA vs ImageNet-L1 U-Net + TTA (matched)	Pearson r	0.7681	0.7678	+0.0003 (-0.0194, +0.0133)	2 / 0 of 3	—
ChipStain + TTA vs ImageNet-L1 U-Net + TTA (matched)	SSIM	0.8137	0.8271	-0.0134 (-0.0323, +0.0103)	0 / 2 of 3	—
ChipStain + TTA vs ImageNet-L1 U-Net + TTA (matched)	MAE	0.0323	0.0291	+0.0032 (+0.0012, +0.0056)	0 / 2 of 3	—
ChipStain + TTA vs ImageNet-L1 U-Net + TTA (matched)	psnr	24.1148	24.4676	-0.3528 (-0.9085, -0.0132)	0 / 0 of 3	—
ChipStain + TTA vs ImageNet-L1 U-Net + TTA (matched)	nuclei F1	0.7082	0.6898	+0.0184 (-0.0092, +0.0419)	2 / 0 of 3	—
ChipStain + TTA vs ImageNet-L1 U-Net + TTA (matched)	$\rho(\sigma, \text{error})$	0.4689	0.3165	+0.1523 (+0.1057, +0.2010)	3 / 0 of 3	4.4e-06
ChipStain + TTA vs ImageNet-L1 U-Net + TTA (matched)	AUSE	0.1775	0.2477	-0.0702 (-0.0947, -0.0453)	3 / 0 of 3	4.4e-06
ImageNet-L1 U-Net + TTA (matched) vs Scratch U-Net + TTA	$\rho(\sigma, \text{error})$	0.3165	0.2832	+0.0334 (-0.0389, +0.1162)	2 / 0 of 3	0.43
ImageNet-L1 U-Net + TTA (matched) vs Scratch U-Net + TTA	AUSE	0.2477	0.2582	-0.0105 (-0.0402, +0.0153)	2 / 0 of 3	0.84
ChipStain vs ImageNet-L1 U-Net + TTA (matched)	$\rho(\sigma, \text{error})$	0.4302	0.3165	+0.1137 (+0.0725, +0.1566)	3 / 0 of 3	6.8e-06
ChipStain vs ImageNet-L1 U-Net + TTA (matched)	AUSE	0.3029	0.2477	+0.0552 (-0.0213, +0.1935)	0 / 0 of 3	1
ChipStain vs Scratch U-Net + TTA	$\rho(\sigma, \text{error})$	0.4302	0.2832	+0.1470 (+0.0900, +0.2271)	3 / 0 of 3	4.4e-06
ChipStain vs Scratch U-Net + TTA	AUSE	0.3029	0.2582	+0.0447 (-0.0344, +0.1813)	0 / 0 of 3	1

Table 2. Paired comparisons; the matched control is the ImageNet-L1 U-Net + TTA. All with TTA on both sides except the single-pass ChipStain rows. Inference rests on the hierarchical seed+field bootstrap CI and on per-seed tests; the pooled field-level p-value averages over seeds, so it is shown for the uncertainty metrics only. For MAE and AUSE lower is better.

6.3 Ablation: what each design choice does

Model	Pearson r $\uparrow$	SSIM $\uparrow$	MAE $\downarrow$	seg-F1 $\uparrow$	$\rho(\sigma, \text{err})$ $\uparrow$	AUSE $\downarrow$
Scratch U-Net (L1, lr 1e-3)	0.749 $\pm$ 0.027	0.826 $\pm$ 0.003	0.0299 $\pm$ 0.0006	0.648 $\pm$ 0.089	—	—
Scratch U-Net (L1, lr 5e-4)	0.724 $\pm$ 0.013	0.815 $\pm$ 0.002	0.0310 $\pm$ 0.0000	0.612 $\pm$ 0.035	—	—
ImageNet-L1 U-Net (L1, lr 5e-4)	0.755 $\pm$ 0.005	0.823 $\pm$ 0.005	0.0296 $\pm$ 0.0003	0.669 $\pm$ 0.021	—	—
ImageNet U-Net (MSE, lr 5e-4)	0.759 $\pm$ 0.042	0.799 $\pm$ 0.008	0.0327 $\pm$ 0.0028	0.685 $\pm$ 0.031	—	—
ImageNet-L1 U-Net + TTA (σ = view s.d.) — matched control	0.768 $\pm$ 0.006	0.827 $\pm$ 0.005	0.0291 $\pm$ 0.0003	0.690 $\pm$ 0.019	0.317 $\pm$ 0.037	0.248 $\pm$ 0.011
ChipStain head, $\beta = 0$ (plain NLL)	0.708 $\pm$ 0.009	0.780 $\pm$ 0.008	0.0348 $\pm$ 0.0009	0.614 $\pm$ 0.019	0.517 $\pm$ 0.032	0.173 $\pm$ 0.022
ChipStain head, $\beta = 0$, + TTA	0.731 $\pm$ 0.017	0.798 $\pm$ 0.002	0.0342 $\pm$ 0.0017	0.640 $\pm$ 0.014	0.520 $\pm$ 0.013	0.160 $\pm$ 0.005
ChipStain head, $\beta = 1$	0.746 $\pm$ 0.013	0.755 $\pm$ 0.029	0.0345 $\pm$ 0.0017	0.671 $\pm$ 0.011	0.350 $\pm$ 0.074	0.315 $\pm$ 0.028
ChipStain head, $\beta = 1$, + TTA	0.768 $\pm$ 0.006	0.793 $\pm$ 0.013	0.0326 $\pm$ 0.0011	0.695 $\pm$ 0.007	0.384 $\pm$ 0.076	0.245 $\pm$ 0.036
ChipStain (β -NLL, $\beta = 0.5$)	0.757 $\pm$ 0.015	0.810 $\pm$ 0.018	0.0324 $\pm$ 0.0013	0.697 $\pm$ 0.012	0.430 $\pm$ 0.006	0.303 $\pm$ 0.072

step (A vs B)	Δ Pearson					
	r	SSIM	MAE	seg-F1	$\rho(\sigma, \text{err})$	AUSE
Learning rate only (scratch U-Net, 5e-4 vs 1e-3)	-0.0251	-0.0114 ▼	+0.0011 ▼	-0.0359	—	—
ImageNet pre-training only (same LR)	+0.0306 ▲	+0.0080	-0.0014 ▲	+0.0571 ▲	—	—
Loss only: L1 $\rightarrow$ MSE (pretrained, single output)	+0.0044	-0.0240 ▼	+0.0031 ▼	+0.0164	—	—
Variance head only: β -NLL vs MSE	-0.0019	+0.0110	-0.0003	+0.0119	—	—
$\beta = 0.5$ vs plain NLL ($\beta = 0$)	+0.0491 ▲	+0.0298 ▲	-0.0024 ▲	+0.0837 ▲	-0.0868 ▼	+0.1302 ▼
$\beta = 0.5$ vs $\beta = 1$	+0.0106	+0.0544 ▲	-0.0021	+0.0261 ▲	+0.0806 ▲	-0.0119

Table 3. Each row changes one factor (A – B; 3 seeds, no TTA). ▲ / ▼ = A robustly better / worse: the seed+field bootstrap CI excludes zero, at least two seeds are significant in that direction and none in the other. For MAE and AUSE lower is better. — = not applicable (no σ) or not tested.

Table 3 changes one factor at a time (3 seeds each, no TTA). Robust effects — the CI excludes zero, at least two seeds are significant in that direction and none in the other — both improvements and degradations: lowering the learning rate to 5e-4 (scratch U-Net): worse SSIM -0.011, MAE +0.001 · ImageNet pre-training (same LR): better Pearson r +0.031, MAE -0.001, nuclei F1 +0.057 · switching the loss from L1 to MSE: worse SSIM -0.024, MAE +0.003 · $\beta = 0.5$ instead of plain NLL ($\beta = 0$): better Pearson r +0.049, SSIM +0.030, MAE -0.002, nuclei F1 +0.084; worse $\rho(\sigma, \text{error})$ -0.087, AUSE +0.130 · $\beta = 0.5$ instead of $\beta = 1$: better SSIM +0.054, nuclei F1 +0.026, $\rho(\sigma, \text{error})$ +0.081 · no robust effect: adding the variance head (β -NLL vs MSE). Effects not listed are within run-to-run variation. Three conclusions follow. (i) The pixel-fidelity cost of ChipStain comes from the change of **loss family**: switching L1 to MSE robustly lowers SSIM and raises MAE, whereas adding the variance head (β -NLL vs MSE) has no robust effect. (ii) β trades fidelity against uncertainty: plain NLL ($\beta = 0$) ranks errors best but loses substantial fidelity (Pearson r 0.708, nuclei F1 0.614); $\beta = 1$ keeps fidelity but loses ranking; $\beta = 0.5$ sits between. We fixed $\beta = 0.5$ before running the ablation, following the recommendation of the β -NLL authors [8]; the ablation shows it is a compromise, not the best choice on every axis. (iii) ImageNet pre-training robustly helps at the same learning rate, while lowering the learning rate alone hurts the scratch model.

6.4 Qualitative results and failure cases

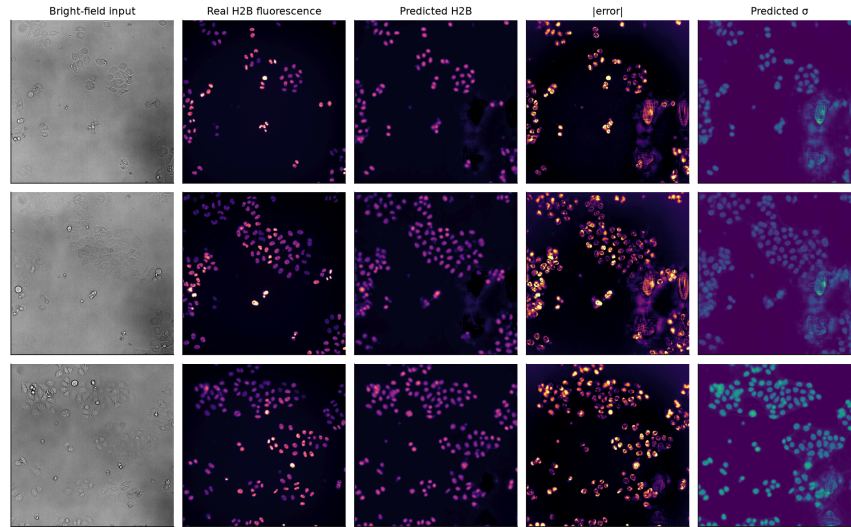
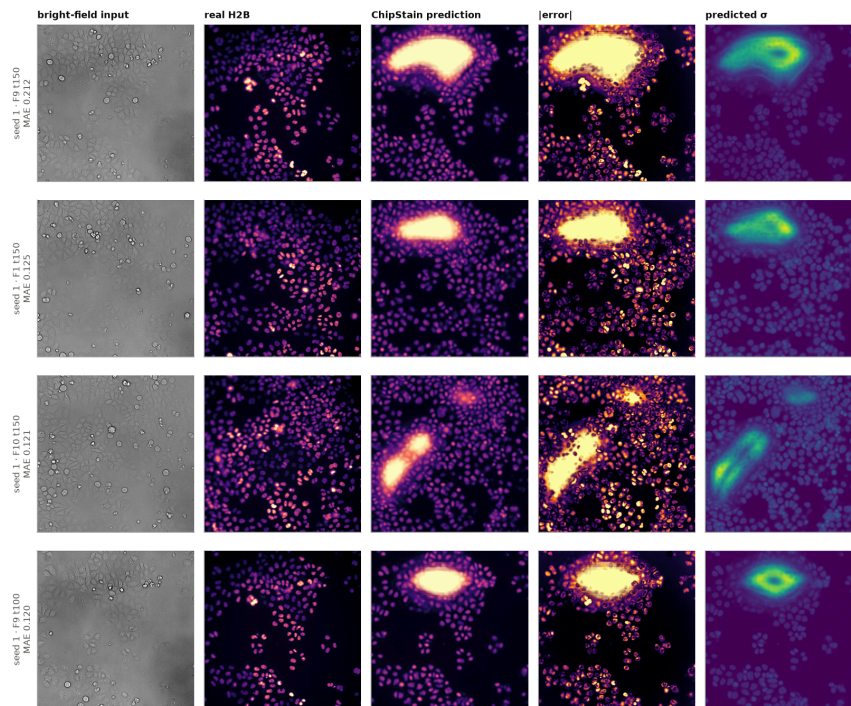


Figure 3. Test images (seed-0 checkpoint, TTA). Columns: bright-field input, real H2B, prediction, absolute error, predicted σ . σ is raised over every nucleus (heteroscedastic σ scales with intensity) and is near zero on background. The strongest σ and error hotspots are the elliptical, grid-textured patches on the right of rows 1–2 (the same field at two time-points, same position) and at the bottom right of row 3: a fixed optical/plate artefact that the model renders as dark textured regions. We did not annotate mitoses, debris or focus; nucleus-level σ is quantified in §6.9. Data: R. Guet, EPFL BIOP, Zenodo 10.5281/zenodo.6140064, CC BY 4.0.



Data: R. Guet, EPFL BIOP, Zenodo 10.5281/zenodo.6140064, CC BY 4.0

Figure 4. The worst test images by MAE across all ChipStain + TTA seeds — all from one training seed (seed 1) at the densest time-point, where that run hallucinates large bright regions. The error is dominated by these regions, and σ is elevated over and around them; these are also that seed's highest- σ test images (§6.10).

6.5 Does σ find the errors better than simple alternatives?

An uncertainty map is only useful if it beats what a user could compute without it. Errors concentrate on bright nuclei and their boundaries, so we rank pixels by σ and, as uncertainty-free proxies, by the predicted intensity μ and its edge strength $|\nabla\mu|$ (Table 4).

Model	$\rho: \sigma$	$\rho: \text{intensity}$	$\rho: \text{edges}$	AUSE: σ	AUSE: intensity	AUSE: edges	gain: σ	gain: intensity	gain: edges	gain: oracle
Scratch U-Net + TTA ($\sigma = \text{view s.d.}$)	0.283 $\pm$ 0.071	0.182 $\pm$ 0.025	0.297 $\pm$ 0.045	0.258 $\pm$ 0.025	0.310 $\pm$ 0.009	0.256 $\pm$ 0.017	40.5 $\pm$ 3.2 %	39.2 $\pm$ 1.8 %	40.0 $\pm$ 2.5 %	55.1 $\pm$ 2.5 %
ImageNet-L1 U-Net + TTA ($\sigma = \text{view s.d.}$) — matched control	0.317 $\pm$ 0.037	0.185 $\pm$ 0.033	0.297 $\pm$ 0.032	0.248 $\pm$ 0.011	0.379 $\pm$ 0.121	0.324 $\pm$ 0.124	42.9 $\pm$ 1.8 %	39.4 $\pm$ 3.7 %	39.5 $\pm$ 3.3 %	56.6 $\pm$ 1.6 %
ChipStain head, $\beta = 0$ (plain NLL)	0.517 $\pm$ 0.032	0.293 $\pm$ 0.048	0.447 $\pm$ 0.027	0.173 $\pm$ 0.022	0.431 $\pm$ 0.107	0.277 $\pm$ 0.120	50.2 $\pm$ 3.8 %	45.1 $\pm$ 3.1 %	44.2 $\pm$ 1.8 %	59.5 $\pm$ 3.1 %
ChipStain head, $\beta = 0$, + TTA	0.520 $\pm$ 0.013	0.295 $\pm$ 0.074	0.441 $\pm$ 0.038	0.160 $\pm$ 0.005	0.344 $\pm$ 0.116	0.274 $\pm$ 0.119	48.6 $\pm$ 3.5 %	44.4 $\pm$ 3.7 %	42.5 $\pm$ 2.2 %	57.6 $\pm$ 2.8 %
ChipStain head, $\beta = 1$	0.350 $\pm$ 0.074	0.484 $\pm$ 0.036	0.374 $\pm$ 0.011	0.315 $\pm$ 0.028	0.303 $\pm$ 0.039	0.283 $\pm$ 0.036	41.8 $\pm$ 1.4 %	44.0 $\pm$ 0.9 %	39.4 $\pm$ 2.2 %	54.5 $\pm$ 0.6 %
ChipStain head, $\beta = 1$, + TTA	0.384 $\pm$ 0.076	0.410 $\pm$ 0.007	0.374 $\pm$ 0.010	0.245 $\pm$ 0.036	0.341 $\pm$ 0.081	0.329 $\pm$ 0.074	45.2 $\pm$ 1.3 %	42.8 $\pm$ 1.2 %	39.6 $\pm$ 1.5 %	55.1 $\pm$ 1.0 %
ChipStain (β -NLL, $\beta = 0.5$)	0.430 $\pm$ 0.006	0.306 $\pm$ 0.020	0.395 $\pm$ 0.017	0.303 $\pm$ 0.072	0.296 $\pm$ 0.043	0.280 $\pm$ 0.087	45.6 $\pm$ 1.6 %	47.8 $\pm$ 4.6 %	44.3 $\pm$ 1.7 %	59.3 $\pm$ 3.4 %
ChipStain + TTA (full)	0.469 $\pm$ 0.027	0.320 $\pm$ 0.041	0.414 $\pm$ 0.030	0.177 $\pm$ 0.021	0.308 $\pm$ 0.104	0.271 $\pm$ 0.060	49.7 $\pm$ 4.2 %	47.7 $\pm$ 5.4 %	43.4 $\pm$ 3.0 %	58.5 $\pm$ 3.7 %

Table 4. ρ : Spearman correlation with |error|. AUSE: area between the sparsification curve and the oracle curve (lower is better). gain: MAE reduction after removing the 20 % highest-score pixels; oracle = removing the 20 % largest true errors. Mean $\pm$ s.d. over seeds, all 125 test images.

Removing the 20 % highest- σ pixels lowers the remaining MAE by 50 % on average over all 125 test images (oracle 59 %), but ranking by edge strength gives 43 % and by intensity 48 %: at that single cut-off σ adds little over a trivial proxy. The full ranking curve tells a different story: AUSE is 0.177 for σ versus 0.271 for edge strength and 0.308 for intensity — σ orders the errors much more faithfully over the whole range (Figure 5).

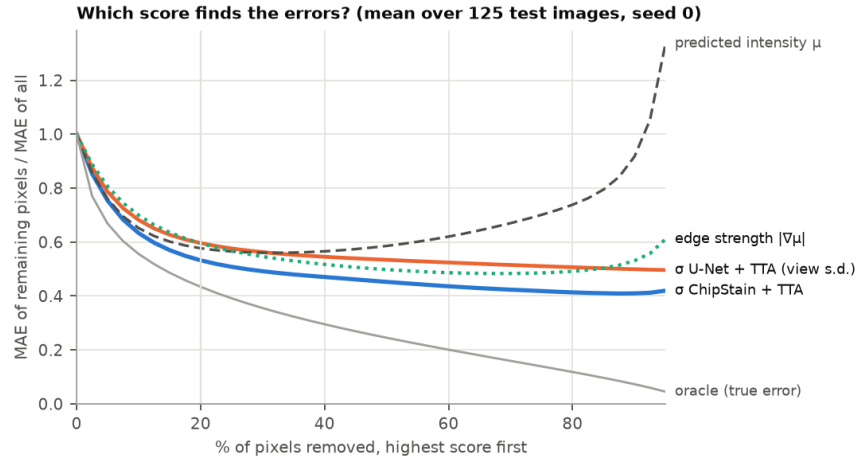


Figure 5. Sparsification curves averaged over all 125 test images (seed 0): MAE of the remaining pixels as pixels are removed in order of each score. Lower is better; the oracle removes the true largest errors first.

Calibration. Ranking is not calibration. Table 5 reports the empirical coverage of $\mu \pm z\sigma$; one scalar fitted on the validation split rescales σ .

model	σ -scale (val)	nominal 50 %	nominal 68.3 %	nominal 80 %	nominal 90 %	nominal 95 %
ChipStain + TTA, raw	—	52.4 %	69.6 %	79.4 %	87.5 %	91.9 %
ChipStain + TTA, recalibrated	$\times 0.98$	51.3 %	68.6 %	78.5 %	86.8 %	91.3 %
ChipStain + TTA, raw, nuclei pixels only	—	56.1 %	73.9 %	83.6 %	90.5 %	93.7 %
ChipStain, single pass, raw	—	45.6 %	63.3 %	74.4 %	84.1 %	89.4 %
ChipStain, single pass, raw, nuclei pixels only	—	52.6 %	69.9 %	79.7 %	87.0 %	90.6 %
ImageNet-L1 U-Net + TTA (view s.d.), raw	—	6.0 %	8.8 %	11.2 %	14.2 %	16.6 %
ImageNet-L1 U-Net + TTA (view s.d.), recalibrated	$\times 18.45$	62.7 %	75.8 %	83.0 %	89.1 %	92.5 %
ImageNet-L1 U-Net + TTA (view s.d.), raw, nuclei pixels only	—	11.7 %	17.2 %	21.7 %	27.0 %	31.4 %
Scratch U-Net + TTA (view s.d.), raw	—	4.9 %	7.4 %	9.6 %	12.4 %	14.8 %
Scratch U-Net + TTA (view s.d.), recalibrated	$\times 21.27$	66.7 %	79.5 %	86.1 %	91.4 %	94.2 %
Scratch U-Net + TTA (view s.d.), raw, nuclei pixels only	—	11.7 %	17.1 %	21.5 %	26.8 %	31.2 %

Table 5. Empirical coverage of the nominal Gaussian intervals on the 125 test images, mean over seeds.

ChipStain + TTA is close to calibrated without any correction: the nominal 68.3 % and 95 % intervals cover 70 % and 91.9 % of test pixels; the scale fitted on the validation split would be $\times 0.98$ (after it, the 95 % interval covers 91.3 %). Coverage is pooled over pixels, most of which are background; restricted to pixels inside the reference nuclei, the 95 % interval covers 94 %. The TTA disagreement of the L1 U-Nets is far too small to be read as a standard deviation (the matched ImageNet U-Net's 68.3 % interval covers 9 %; it needs scaling by $\times 18$, the scratch U-Net by $\times 21$): it can rank errors but not quantify them. Calibration holds in distribution only: under imaging shift (§6.7) σ does not track the much larger error in most training runs.

6.6 Image level: is mean σ more than a density counter?

Model	$\rho(\text{mean } \sigma, \text{MAE})$	$\rho(n_{\text{pred}}, \text{MAE})$	partial $\rho(\sigma, \text{MAE})$ given n_{pred}	$\rho(\text{mean } \sigma, 1-F1)$	$\rho(n_{\text{pred}}, 1-F1)$	$\rho(\text{mean } \sigma, 1-r)$
Scratch U-Net + TTA ($\sigma = \text{view s.d.}$)	0.903 $\pm$ 0.029	0.872 $\pm$ 0.021	0.515 $\pm$ 0.108	0.406 $\pm$ 0.044	0.292 $\pm$ 0.060	0.648 $\pm$ 0.022
ImageNet-L1 U-Net + TTA ($\sigma = \text{view s.d.}$) — matched control	0.771 $\pm$ 0.275	0.896 $\pm$ 0.022	0.462 $\pm$ 0.218	0.310 $\pm$ 0.081	0.276 $\pm$ 0.035	0.586 $\pm$ 0.049
ChipStain head, $\beta = 0$ (plain NLL)	0.945 $\pm$ 0.016	0.939 $\pm$ 0.015	0.373 $\pm$ 0.045	0.317 $\pm$ 0.097	0.279 $\pm$ 0.104	0.491 $\pm$ 0.162
ChipStain head, $\beta = 0$, + TTA	0.936 $\pm$ 0.016	0.929 $\pm$ 0.013	0.358 $\pm$ 0.115	0.374 $\pm$ 0.067	0.331 $\pm$ 0.078	0.575 $\pm$ 0.087
ChipStain head, $\beta = 1$	0.942 $\pm$ 0.022	0.940 $\pm$ 0.015	0.374 $\pm$ 0.188	0.379 $\pm$ 0.085	0.337 $\pm$ 0.043	0.481 $\pm$ 0.183
ChipStain head, $\beta = 1$, + TTA	0.882 $\pm$ 0.040	0.933 $\pm$ 0.014	0.308 $\pm$ 0.190	0.333 $\pm$ 0.081	0.321 $\pm$ 0.033	0.569 $\pm$ 0.074
ChipStain (β -NLL, $\beta = 0.5$)	0.948 $\pm$ 0.022	0.907 $\pm$ 0.031	0.596 $\pm$ 0.233	0.392 $\pm$ 0.108	0.308 $\pm$ 0.037	0.622 $\pm$ 0.072
ChipStain + TTA (full)	0.908 $\pm$ 0.067	0.891 $\pm$ 0.028	0.628 $\pm$ 0.290	0.426 $\pm$ 0.135	0.303 $\pm$ 0.034	0.650 $\pm$ 0.089

Table 6. Per-image Spearman correlations over the 125 test images (mean $\pm$ s.d. over seeds). n_{pred} = number of nuclei detected on the prediction (needs no uncertainty).

Mean σ ranks test images by their MAE ($\rho = 0.91$), but in-distribution most of this is cell density: the predicted nucleus count alone reaches $\rho = 0.89$, and the U-Net's TTA disagreement reaches 0.90. Controlling for the count, mean σ still carries independent information (partial $\rho = 0.63 \pm 0.29$ over seeds), but so does the U-Net's TTA disagreement (partial $\rho = 0.52$; $\rho(\bar{\sigma}, 1 - r)$ 0.65 vs 0.65 for ChipStain). In distribution, the frame-level ranking is therefore **not** an advantage of the β -NLL head. Where frame-level σ matters is when a run or the input goes wrong: one ChipStain seed's three catastrophic test frames are its three highest- σ images (§6.10), and under imaging shift (§6.7) density proxies do not change while the error does.

6.7 Imaging shift: does the uncertainty notice?

Moving from a well plate to a chip can change the optics (thicker PDMS, curved channel walls, scattering), and labs differ in camera noise and modality. We applied simulated shifts to 50 test images (time-points 10 and 100), for three training seeds of each model: Gaussian blur of 1, 2 and 4 px (a crude defocus proxy), added sensor noise (10 % and 20 %), and a modality swap to the dataset's digital phase-contrast rendering of the same fields.

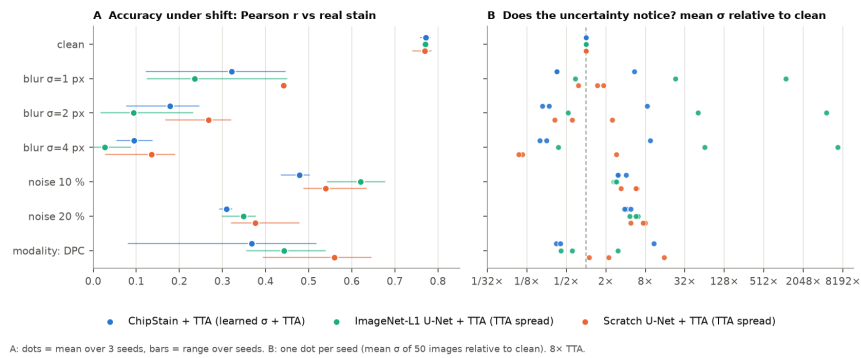


Figure 6. (A) Pearson r with the real stain under each shift. (B) Mean σ relative to the clean image (log scale). Dots are means over three seeds and thin bars the range over seeds; a mean above $1\times$ can hide seeds in which σ falls (Table 7b). All models use $8\times$ TTA; for the L1 U-Nets σ is the TTA disagreement.

model	σ term	blur	noise	modality (DPC)
U-Net + TTA	total σ	0.43 ± 0.41	1.00 ± 0.00	0.85 ± 0.20
U-Net + TTA	TTA disagreement	0.43 ± 0.41	1.00 ± 0.00	0.85 ± 0.20
ChipStain + TTA	total σ	0.34 ± 0.57	1.00 ± 0.00	0.33 ± 0.58
ChipStain + TTA	learned head (TTA mean)	0.30 ± 0.51	1.00 ± 0.00	0.32 ± 0.56
ChipStain + TTA	TTA disagreement	0.35 ± 0.57	0.93 ± 0.04	0.39 ± 0.53
ChipStain + TTA	learned head, single pass	0.26 ± 0.43	1.00 ± 0.00	0.30 ± 0.52
+ ImageNet encoder + TTA	total σ	0.71 ± 0.49	0.98 ± 0.01	0.42 ± 0.47
+ ImageNet encoder + TTA	TTA disagreement	0.71 ± 0.49	0.98 ± 0.01	0.42 ± 0.47

Table 7a. Shift detection: AUROC of per-image mean σ for separating shifted from clean images, per uncertainty term, mean $\pm$ s.d. over three seeds ($1 =$ perfect separation, $0.5 =$ no signal, $< 0.5 = \sigma$ falls under shift).

model	σ term	failure AUROC	failed images	within-condition $p(\sigma, \text{MAE})$
U-Net + TTA	total σ	0.61 ± 0.25	640	0.42 ± 0.27
U-Net + TTA	TTA disagreement	0.61 ± 0.25	640	0.42 ± 0.27
ChipStain + TTA	total σ	0.56 ± 0.38	768	0.69 ± 0.03
ChipStain + TTA	learned head (TTA mean)	0.54 ± 0.35	768	0.43 ± 0.28
ChipStain + TTA	TTA disagreement	0.57 ± 0.36	768	0.51 ± 0.17
ChipStain + TTA	learned head, single pass	0.52 ± 0.30	768	0.42 ± 0.30
+ ImageNet encoder + TTA	total σ	0.74 ± 0.38	695	0.51 ± 0.33
+ ImageNet encoder + TTA	TTA disagreement	0.74 ± 0.38	695	0.51 ± 0.33

Table 7b. Failure detection: AUROC of per-image mean σ for separating failed predictions (Pearson $r < 0.5$) from successful ones ($r > 0.7$), clean and shifted images pooled, mean $\pm$ s.d. over three seeds. 'Failed images' are summed over the three seeds (of 1,050 shifted and clean images per model).

All models' predictions degrade under blur and the modality swap (Figure 6A), and ChipStain is not more robust than the L1 models: under blur it lies between them, with phase contrast it is the worst (mean Pearson r under 1 px blur 0.32 for ChipStain vs 0.24 for the matched ImageNet U-Net and 0.44 for the scratch U-Net; with phase contrast 0.37 vs 0.44 and 0.56). Whether σ then rises is a property of the individual training run rather than of the method (Figure 6B, one dot per seed): mean σ separates blurred from clean images (AUROC > 0.8) in 1 of 3 ChipStain seeds, 2 of 3 ImageNet-U-Net seeds and 1 of 3 scratch-U-Net seeds, and *falls* under blur (AUROC < 0.2 , a silent failure) in 2 of 3, 1 of 3 and 1 of 3 seeds respectively. Averaged over seeds, ChipStain's σ is the **weakest** shift detector of the three (shift-detection AUROC for blur 0.34 vs 0.71 and 0.43; phase contrast 0.33 vs 0.42 and 0.85; failure detection 0.56 vs 0.74 and 0.61; Tables 7a–b). Where a signal appears, it comes from the **TTA disagreement**, not from the learned β -NLL head: the single-pass learned σ has the lowest shift AUROC of all terms (blur 0.26, phase contrast 0.30). The released seed-0 checkpoint is one of the runs whose total σ rises ($\times 9.5$ under 4 px blur, $\times 10.8$ with phase contrast), which is what the demo shows; with the same checkpoint the single-pass learned σ actually falls on the blurred demo image. Additive noise of 10–20 % raises total σ in every model and seed (shift-detection AUROC ≥ 0.98); the validation-calibrated gate flags 97–100 % of noisy images for ChipStain and the scratch U-Net but only 59–67 % for the ImageNet-L1 U-Net. Affine contrast changes were not tested, because the per-image normalisation cancels them exactly. **Conclusion:** in distribution ChipStain's σ ranks errors well and is calibrated (§6.5); out of distribution neither its σ nor TTA disagreement is a dependable drift alarm. An input check does this job better (below).

A cheap input check catches what σ misses

Because σ cannot be trusted to notice a shift, we tested two checks on the *input* that need no retraining, on the same 50 images, shifts and three seeds (`scripts/input_qc.py`). The **focus check** is model-free: the log variance of the Laplacian of the normalised bright-field image, scored as its distance from the training median (blur lowers it, noise raises it). The **feature check** is the Mahalanobis distance of the ChipStain encoder's deepest feature map (global average pool, 512 dimensions, Ledoit-Wolf covariance from the 200 training images). Each threshold is the 95th percentile of the score on the clean validation images, the same rule as the σ gate. Both checks separate blurred and phase-contrast images from clean ones almost perfectly (shift-detection AUROC: focus 1.00 and 1.00, features 1.00 and 0.99, against 0.34 and 0.33 for total σ ; Table 7c), and both flag 100 % of images blurred by only 1 px and 100 % of phase-contrast images in every seed (Table 7d). They also predict failure far better than σ (failure AUROC 0.95 and 0.99 vs 0.56), although with clean and shifted images pooled this mostly reflects that the shifted images are the ones that fail. The checks are complementary to σ : the focus check misses mild noise (it flags 32 % of images with 10 % noise), which σ catches, and the feature check is seed-dependent on noise. Combining the focus check with the σ gate flags 100 % of blurred, 98 % of noisy (10 %; at least 94 % in every seed) and 100 % of phase-contrast images. The cost is false alarms. On clean test images the focus check flags 16 % (the combination 17 %), against 5 % expected on validation. These are fields of the held-out well whose focus differs slightly from the training wells, and their predictions are good (mean r 0.81). They cost review time, not missed failures, and they show that the focus threshold should be re-set on a few clean images of each new plate or device. **Recommended use:** σ for trust inside the training domain, and a cheap input check for drift. These are simulated shifts; real chip optics remain untested.

check	blur	noise	modality (DPC)	failure AUROC ($r < 0.5$ vs $r > 0.7$)
focus (variance of Laplacian)	1.00 $\pm$ 0.00	0.84 $\pm$ 0.00	1.00 $\pm$ 0.00	0.95 $\pm$ 0.01
encoder features (Mahalanobis)	1.00 $\pm$ 0.00	0.98 $\pm$ 0.03	0.99 $\pm$ 0.01	0.99 $\pm$ 0.01
total σ (for comparison)	0.34 $\pm$ 0.57	1.00 $\pm$ 0.00	0.33 $\pm$ 0.58	0.56 $\pm$ 0.38

Table 7c. Input-level checks on the same images and seeds (ChipStain encoder): AUROC for separating shifted from clean images, and for separating failed ($r < 0.5$) from successful ($r > 0.7$) predictions, clean and shifted pooled; mean $\pm$ s.d. over three seeds. Total σ is repeated from Tables 7a–b for comparison. The focus check does not depend on the model, so its s.d. is zero for shift detection.

check	clean	blur $\sigma=1$ px	blur $\sigma=2$ px	blur $\sigma=4$ px	noise 10 %	noise 20 %	modality: DPC
focus	16 %	100 %	100 %	100 %	32 %	100 %	100 %
encoder features	9 % (0–22)	100 %	100 %	100 %	67 % (0–100)	67 % (0–100)	100 %
either	23 % (16–34)	100 %	100 %	100 %	77 % (32–100)	100 %	100 %
focus or σ gate	17 % (16–20)	100 %	100 %	100 %	98 % (94–100)	100 %	100 %

Table 7d. Fraction of images flagged per condition when each check's threshold is the 95th percentile of its score on the clean validation images; mean over seeds (range in brackets where seeds differ). 'Focus or σ gate' combines the focus check with the σ gate of the shift test. This combination was chosen after seeing the per-check results.

6.8 Downstream counting and the direct-segmentation alternative

Counting nuclei on the prediction with a fixed, simple watershed reaches F1 0.708 (ChipStain + TTA) against the StarDist reference; the same watershed on the real stain reaches 0.765, so much of the gap to perfect agreement is the segmenter (Appendix A). Count agreement is better than object agreement because merges and splits partly cancel. A biologist will ask whether one should predict the stain at all if the goal is a count. The dataset author's Cellpose model, trained on this dataset's training split to segment nuclei directly from bright-field (and quality-controlled by its author on this held-out well), reaches F1 **0.831** on the same test images with a mean count error of 6.4 % (Table 8) — better than counting on our prediction, and better than our watershed on the real stain, because it was trained to reproduce exactly the StarDist label definition used as reference. **For counting alone, direct segmentation is the better tool.** Replacing our watershed by Cellpose's generic fluorescence nuclei model, F1 on the real stain is 0.846 and on ChipStain's predictions 0.674 — part of the gap is the segmenter, part is the prediction. ISL earns its place differently: it returns an intensity image that any downstream pipeline can use (intensity, texture, chromatin condensation, morphology — not only outlines), and a calibrated uncertainty. Under 2 px blur Cellpose's F1 drops from 0.83 to 0.00 and it returns no uncertainty at all; ChipStain's σ does not reliably warn in that situation either (§6.7), so neither route protects a user against an imaging change. We also tested whether ChipStain's mean σ predicts where Cellpose fails in distribution; it does not ($p = 0.05$), so we make no claim that the two are complementary there.

time-point	F1	n_pred	n_ref
1	0.871	71	73
10	0.866	77	78
50	0.834	104	104
100	0.799	152	157
150	0.784	209	222

Table 8. Cellpose bright-field nuclei model published with the dataset, per time-point, against the same StarDist references.

6.9 Per-nucleus uncertainty

model	score	pooled AUROC	within-image AUROC	seeds
ChipStain + TTA	dim	0.655 ± 0.031	0.650	3
ChipStain + TTA	sigma_learned	0.627 ± 0.024	0.710	3
ChipStain + TTA	sigma_learned_imgnorm	0.665 ± 0.032	0.710	3
ChipStain + TTA	sigma_learned_rel	0.769 ± 0.043	0.778	3
ChipStain + TTA	small	0.546 ± 0.039	0.524	3
ChipStain ensemble (3 seeds + TTA)	dim	0.653	0.640	1
ChipStain ensemble (3 seeds + TTA)	sigma_ensemble	0.620	0.719	1
ChipStain ensemble (3 seeds + TTA)	sigma_ensemble_imgnorm	0.666	0.719	1
ChipStain ensemble (3 seeds + TTA)	sigma_ensemble_rel	0.776	0.783	1
ChipStain ensemble (3 seeds + TTA)	small	0.528	0.509	1
U-Net + TTA	dim	0.743 ± 0.058	0.721	3
U-Net + TTA	sigma_tta_unet	0.721 ± 0.011	0.732	3
U-Net + TTA	sigma_tta_unet_imgnorm	0.721 ± 0.023	0.732	3
U-Net + TTA	sigma_tta_unet_rel	0.773 ± 0.004	0.759	3
U-Net + TTA	small	0.682 ± 0.066	0.672	3

Table 9. AUROC of each per-nucleus score for flagging predicted nuclei that match no reference nucleus (IoU < 0.5). For ChipStain + TTA, 'sigma_learned' is the total σ (learned variance + TTA disagreement); for the L1 U-Net, 'sigma_tta_unet' is its TTA disagreement.

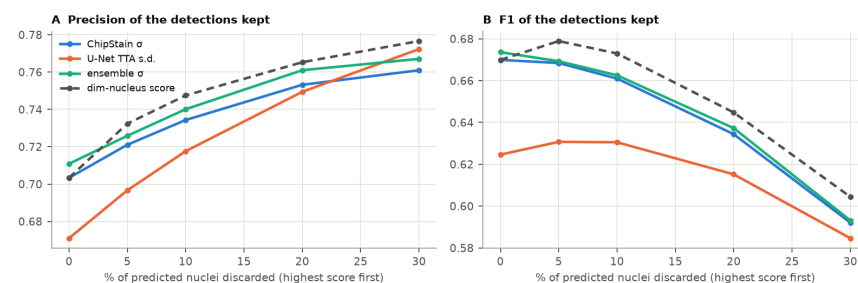


Figure 7. Discarding the highest-score detections raises precision but lowers recall and F1 for every score: per-nucleus σ is a review flag, not a count filter.

At the level of single nuclei, σ is weak. The mean σ inside a predicted nucleus flags detections that match no reference nucleus with AUROC 0.63 — slightly below a trivial 'dim nucleus' score on the same predictions (0.66) — and the U-Net's TTA disagreement does better on its own predictions (0.72); the 3-seed ensemble does not help (0.62). Discarding uncertain detections raises precision but lowers recall and F1 for every score (Figure 7). We therefore use per-nucleus σ only as a visual review aid and claim no advantage at this level.

6.10 A biological read-out: proliferation from bright-field alone

The organisers ask for predictions that enable downstream analysis. The most direct biological quantity in this time-lapse is the population doubling time. Per test field we fit $\log(\text{count})$ against time over the five time-points (0–37 h) and compare doubling times from label-free counts with those from the StarDist reference annotation of the real stain.

model	seed	estimate	fields used	doubling time (h)	mean signed bias	mean abs. error	collapsed fields	fields to review	frames flagged	failed frames (F1 < 0.5) caught
Scratch U-Net + TTA	0	all frames	25	24.9	+7.7 %	11.3 %	0	—	—	—
Scratch U-Net + TTA	0	all frames, same fields as gated	19	24.3	+4.3 %	9.0 %	—	—	—	—
Scratch U-Net + TTA	0	σ -gated	19	24.3	+4.1 %	8.9 %	—	6	26	4/6
Scratch U-Net + TTA	1	all frames	25	24.7	+6.8 %	10.7 %	0	—	—	—
Scratch U-Net + TTA	1	all frames, same fields as gated	21	24.2	+4.2 %	8.8 %	—	—	—	—
Scratch U-Net + TTA	1	σ -gated	21	24.0	+3.6 %	8.5 %	—	4	22	4/4
Scratch U-Net + TTA	2	all frames	25	29.0	+26.7 %	28.6 %	0	—	—	—
Scratch U-Net + TTA	2	all frames, same fields as gated	20	27.6	+19.4 %	21.6 %	—	—	—	—
Scratch U-Net + TTA	2	σ -gated	20	27.5	+18.7 %	21.0 %	—	5	24	8/46
ChipStain + TTA	0	all frames	25	23.8	+3.2 %	9.2 %	0	—	—	—
ChipStain + TTA	0	all frames, same fields as gated	22	23.5	+1.3 %	8.1 %	—	—	—	—
ChipStain + TTA	0	σ -gated	22	23.5	+1.5 %	8.6 %	—	3	15	0/3
ChipStain + TTA	1	all frames	23	26.1	+15.1 %	19.6 %	2	—	—	—
ChipStain + TTA	1	all frames, same fields as gated	20	25.0	+7.9 %	13.1 %	—	—	—	—
ChipStain + TTA	1	σ -gated	20	24.8	+7.1 %	12.9 %	—	5	29	8/9
ChipStain + TTA	2	all frames	25	25.1	+9.1 %	14.0 %	0	—	—	—
ChipStain + TTA	2	all frames, same fields as gated	23	24.9	+8.2 %	13.6 %	—	—	—	—
ChipStain + TTA	2	σ -gated	23	23.9	+3.7 %	10.0 %	—	2	16	1/4
ImageNet-L1 U-Net + TTA	0	all frames	25	25.2	+9.0 %	12.4 %	0	—	—	—
ImageNet-L1 U-Net + TTA	0	all frames, same fields as gated	20	24.8	+6.9 %	11.1 %	—	—	—	—
ImageNet-L1 U-Net + TTA	0	σ -gated	20	24.8	+7.0 %	11.2 %	—	5	28	4/12
ImageNet-L1 U-Net + TTA	1	all frames	25	25.2	+8.8 %	12.0 %	0	—	—	—
ImageNet-L1 U-Net + TTA	1	all frames, same fields as gated	18	25.4	+8.9 %	12.6 %	—	—	—	—
ImageNet-L1 U-Net + TTA	1	σ -gated	18	24.7	+5.9 %	9.8 %	—	7	29	4/12
ImageNet-L1 U-Net + TTA	2	all frames	25	26.0	+12.8 %	15.9 %	0	—	—	—
ImageNet-L1 U-Net + TTA	2	all frames, same fields as gated	22	25.3	+9.1 %	12.7 %	—	—	—	—
ImageNet-L1 U-Net + TTA	2	σ -gated	22	25.1	+8.5 %	12.1 %	—	3	14	3/19

Table 10. Doubling time (geometric mean over the fields used) and its error against the reference, per run, three ways: all frames; all frames on the same fields that the gate keeps; and gated (threshold fixed on the validation split, §4.4). 'Collapsed fields' have no positive growth fit on all frames and are excluded from the all-frames row

(counted as failures in the text); 'fields to review' are fields the gate leaves with fewer than three frames. Bias is the mean signed error over fields. 'Failed frames' are test frames whose nuclei F1 is below 0.5.

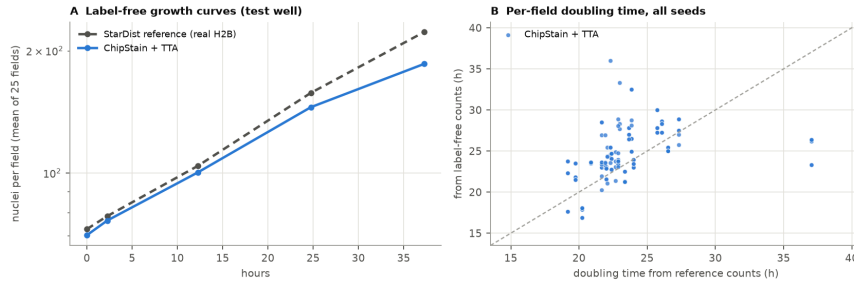


Figure 8. (A) Mean nuclei per field over time from the reference and from label-free counts (seed 0). (B) Per-field doubling times, all seeds.

Label-free counts reproduce the reference doubling time (23.3 h) with a mean signed bias of 9.1 % (mean of the absolute per-run bias over the three ChipStain seeds; 10.2 % for the matched ImageNet U-Net, 13.8 % for the scratch U-Net). The errors are driven by failing fields: in one ChipStain run 2 fields collapse completely (no positive growth fit) and are therefore missing from the all-frames estimate — they count as failures, and that run's bias (15.1 %) understates its problem. The σ gate (threshold fixed on the validation split with a 5 % false-alarm budget; each frame's mean σ is taken relative to the median of the frames at the same time-point in the same run, so the gate needs a batch of frames per time-point) drops flagged frames and routes fields left with fewer than three frames to review. Compared **on the same fields**, most of the gain comes from that routing: ChipStain's mean bias is 9.1 % on all fields, 5.8 % on the fields the gate keeps without removing any frame, and 4.1 % gated (mean absolute per-field error 14.3 $\rightarrow$ 11.6 $\rightarrow$ 10.5 %). The gate helps the L1 models too (matched control 10.2 $\rightarrow$ 8.3 $\rightarrow$ 7.1 %; scratch 13.8 $\rightarrow$ 9.3 $\rightarrow$ 8.8 %), so it is not specific to ChipStain. It flags 60 of 375 ChipStain test frames (16 %, far above the 5 % budget; the flags concentrate on a few fixed fields) and catches 9/16 of the frames whose detection failed ($F1 < 0.5$); it catches catastrophic failures, not moderate ones. In total 10 field-runs are routed to review.

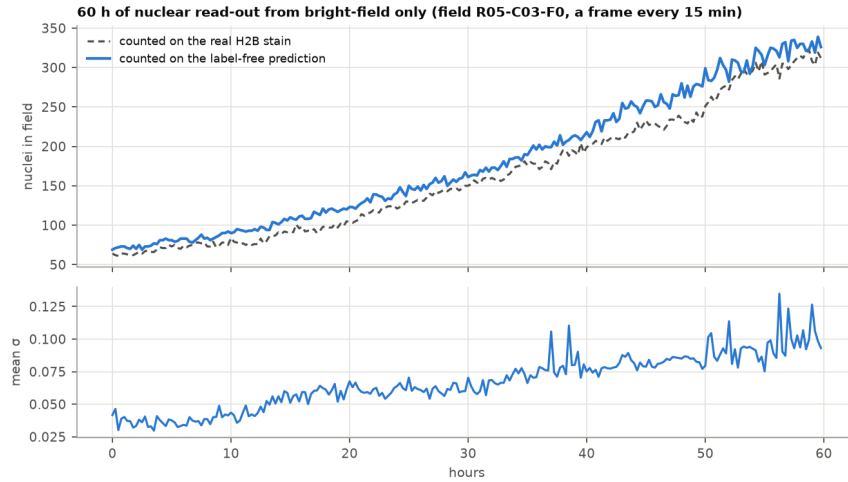
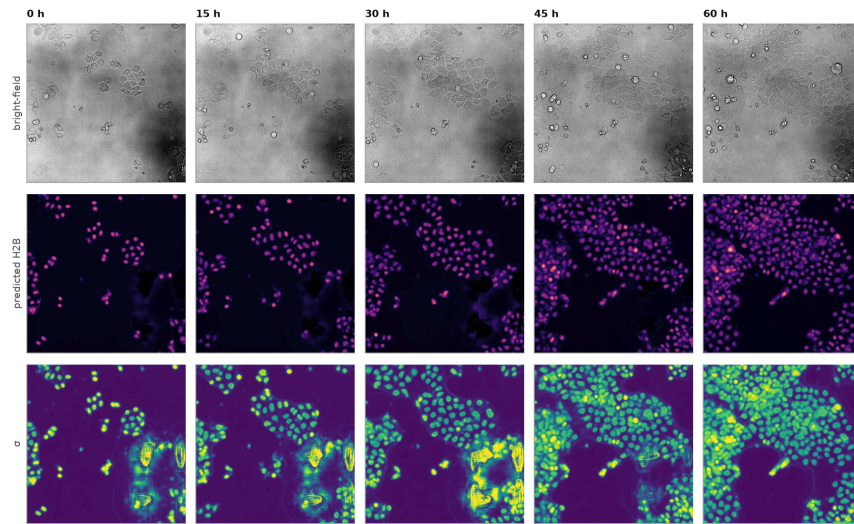


Figure 9. One held-out field imaged every 15 min for 60 h (240 frames): nuclei counted on the label-free prediction and on the real H2B stain with the same pipeline (top), and the frame's mean σ (bottom). Data: R. Guet, EPFL BIOP, Zenodo 10.5281/zenodo.6139958, CC BY 4.0.



Data: R. Guilet, EPFL BIOP, Zenodo 10.5281/zenodo.6139958, CC BY 4.0

Figure 10. Frames at 0, 15, 30, 45 and 60 h: bright-field input, predicted H2B and σ .

On the full recording of one held-out field (240 frames over 60 h, Figures 9–10) the label-free counts follow the real-stain counts with a mean error of 12 % and give a doubling time of 24.9 h versus 23.9 h, both counted with the same watershed – a growth curve from bright-field alone, without a single fluorescence exposure. Object-level agreement between the two segmentations is modest (median F1 0.54) because both use the simple watershed. Mean σ rises as the field becomes crowded and tracks per-frame failure ($\rho(\text{mean } \sigma, 1 - \text{F1}) = 0.87$), which is confounded by density and time; frames after ≈ 37 h are beyond the densest training time-point. Five of the 240 frames are also test images; the other 235 were never used.

6.11 Transfer to neural cultures

We applied the HeLa models to human iPSC-derived motor-neuron cultures (Christiansen et al. 2018, Condition A; 22 training and 3 test wells, bright-field z-stacks and widefield DAPI) – neurons in a dish, not an OoC. The rescale factor (0.55) and the bright-field z-plane (7) were chosen on two validation wells with the ChipStain checkpoint and agree with a nucleus-size estimate computed from the DAPI images of all wells. **Without weight updates** (scale and plane chosen on the two labelled validation wells), ChipStain reaches Pearson r 0.59 with the real DAPI (U-Net 0.53), but counting on the prediction is poor (F1 against the real-DAPI segmentation 0.20; count -48 %), and its mean σ (0.080) stays inside the HeLa test range (0.020–0.101 (5th–95th percentile, with TTA)): the uncertainty does **not** signal that the model is out of domain – consistent with §6.7. **Fine-tuning** from the HeLa weights with a single training well raises r to 0.76 (F1 0.41), and with 20 wells to 0.77 (F1 0.44); starting from ImageNet instead gives 0.74 and 0.74. These fine-tuning runs use one seed and the first k wells, so small differences are not meaningful; the experiment shows that a small paired set adapts the model to a new cell type and imaging system.

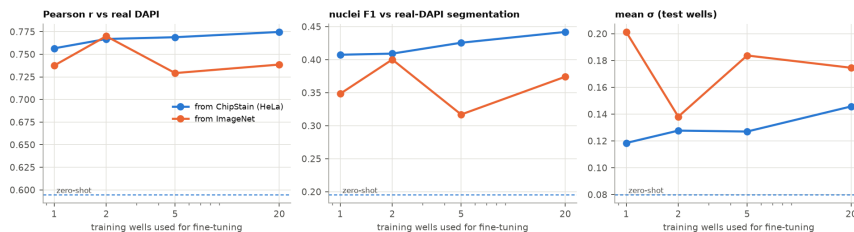
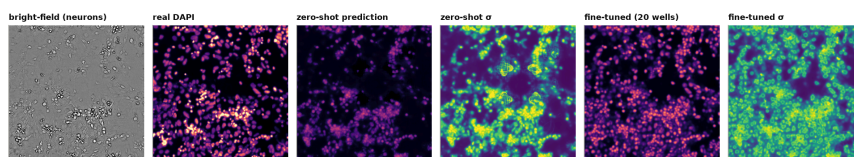


Figure 11. Human iPSC-derived motor neurons (Christiansen et al. 2018, CC BY 4.0), 3 test wells. Fine-tuning with k training wells from the HeLa ChipStain weights (blue) or from ImageNet only (orange); dashed line: HeLa model applied zero-shot.



Data: Christiansen et al., Cell 2018, In-silico-labeling Condition A, CC BY 4.0 (rescaled, normalised, colour-mapped). σ panels share one colour scale.

Figure 12. Neural test well: bright-field, real Hoechst stain, zero-shot prediction and σ , fine-tuned prediction and σ .

6.12 Deep ensembles

Combining the three training seeds, a deep ensemble of plain U-Nets yields a poor uncertainty (ρ 0.12, AUSE 0.372): with three members, the disagreement of point predictors is too noisy to rank errors. An ensemble of ImageNet-initialised L1 U-Nets does better (ρ 0.32, AUSE 0.270) but still trails a single ChipStain model with TTA. A ChipStain ensemble without TTA gives the best ranking among $\beta = 0.5$ configurations (ρ 0.48, AUSE 0.169; the plain-NLL head with TTA ranks errors better still, ρ 0.52, AUSE 0.160, at a large fidelity cost — \$6.3), and with TTA it reaches Pearson r 0.774 and seg-F1 0.714, absorbing single-seed failures. Three models triple the cost, so the single TTA model is the released default.

run	mae	psnr	ssim	pearson	seg_f1	seg_f1_realfluo	spearman_unc_err	ause	gain20	spearman_mu_err	ause_mu
ens3_baseline_unet	0.029 ± 0.012	24.479 ± 2.859	0.833 ± 0.074	0.769 ± 0.058	0.684 ± 0.118	0.765 ± 0.087	0.117 ± 0.281	0.372 ± 0.171	0.372 ± 0.132	0.137 ± 0.241	0.333 ± 0.126
ens3_baseline_unet_tta	0.029 ± 0.012	24.568 ± 2.857	0.835 ± 0.073	0.775 ± 0.056	0.691 ± 0.118	0.765 ± 0.087	0.123 ± 0.294	0.359 ± 0.178	0.387 ± 0.138	0.141 ± 0.240	0.329 ± 0.128
ens3_chipstain_nll	0.032 ± 0.017	24.054 ± 3.364	0.818 ± 0.084	0.765 ± 0.071	0.699 ± 0.128	0.765 ± 0.087	0.475 ± 0.172	0.169 ± 0.061	0.497 ± 0.105	0.296 ± 0.248	0.288 ± 0.117
ens3_chipstain_nll_tta	0.032 ± 0.017	24.235 ± 3.224	0.816 ± 0.080	0.774 ± 0.062	0.714 ± 0.114	0.765 ± 0.087	0.465 ± 0.164	0.180 ± 0.054	0.477 ± 0.096	0.322 ± 0.229	0.298 ± 0.136
ens3_pretrained_l1	0.029 ± 0.012	24.416 ± 2.823	0.823 ± 0.074	0.764 ± 0.054	0.687 ± 0.129	0.765 ± 0.087	0.322 ± 0.126	0.270 ± 0.059	0.424 ± 0.100	0.160 ± 0.225	0.437 ± 0.294

Table 11. 3-seed deep ensembles (all 125 test images): mean of the member predictions; σ^2 = variance of member means + mean member σ^2 .

7. Reliability analysis and limitations

- **No organ-on-a-chip data; one plate, one session.** All validation uses 2-D cells in a well plate (HeLa) or a dish (neurons). The three HeLa wells come from one plate and one imaging session, so the hold-out measures well-to-well generalisation only. The shift test approximates some chip optics, not 3-D tissue, flow or PDMS autofluorescence. Before use on a chip, fine-tune on paired images from that device and re-check Tables 1, 5 and 7.
- **Fidelity cost.** Against the matched ImageNet U-Net with TTA, ChipStain has equal correlation but 11 % higher MAE and lower SSIM; the ablation traces this to the squared-error loss family.
- **Robustness and drift.** Under blur and modality change every model's prediction fails and ChipStain's is not more robust (between the two L1 models under blur, worst with phase contrast). Whether σ then rises is seed-dependent for every model (1 of 3 ChipStain seeds), ChipStain's σ is the weakest of the three drift detectors on average, and any signal comes from the TTA term; σ is not a dependable drift alarm. A cheap input check (focus and encoder-feature distance) catches the simulated blur and modality shifts in every seed, but it was tested only on simulated shifts and flags 16 % of clean images from the new well.
- **Scope of the uncertainty.** σ ranks pixel errors well and is near-calibrated in distribution, but at a single 20 % cut-off it is only slightly better than ranking by predicted intensity; at the nucleus level it is no better than a dim-nucleus heuristic; in distribution its frame-level signal is matched by the U-Net's TTA disagreement and largely reflects cell density; and the frame gate catches catastrophic, not moderate, failures.
- **Counting.** A direct bright-field segmenter trained on matching labels counts nuclei better than counting on our prediction (\$6.8).
- **Automatic references.** F1 and counts are measured against StarDist labels on the real stain, not manual annotation; our watershed on the real stain reaches F1 0.765 — a reference level, not a bound.
- **Training variability.** One of three ChipStain seeds fails on two test fields (σ flags them); one U-Net seed loses 0.16 nuclei F1. Retraining lands within the seed spread of Table 1, not on identical numbers; hence three seeds and seed-aware statistics throughout.
- **Target.** mCherry-H2B is a genetically encoded reporter; on this dataset no dye is actually avoided. The value applies where such a reporter is unavailable.
- **Ethics.** No human-subject, clinical or personal data. The main risk is over-trusting a hallucinated prediction, which σ and the gate are designed to reduce but cannot eliminate.
- **Proliferation gate.** Most of the gated improvement comes from routing failing fields to review; the gate normalises each frame by the other frames at the same time-point, so it needs a batch of frames.

8. Impact and the path to organ-on-a-chip deployment

Where ChipStain could help. Once fine-tuned on a small paired set from the target device, it could provide a quality-controlled, label-free nuclear channel for long recordings: growth and viability curves without fixation or repeated dye exposure, a freed fluorescence channel for a functional reporter, and a σ gate that keeps catastrophic frames out of the analysis (§6.10; the only proliferation evidence so far is from a well plate). Its mean σ should *not* be relied on as an imaging-drift alarm: under blur or a change of modality it rises only in some training runs (§6.7), so drift should be caught by the cheap input check of §6.7 (focus and encoder-feature distance), with its threshold re-set per device, and new imaging conditions confirmed with a few paired images. For pure nuclear counting a direct segmenter should be preferred (§6.8).

Neural OoC (the supporting organisation's focus). Neurons are post-mitotic, so label-free nuclear counts and nuclear condensation over weeks are a natural viability/neurotoxicity endpoint; repeated live DNA dyes perturb such cultures, and reporter lines are rarely available for iPSC-derived neurons. §6.11 tests transfer to human iPSC-derived motor neurons.

Path to a chip. (1) Acquire a small set of paired bright-field / nuclear-stain images on the target device (we did not measure how many are needed on a chip; §6.11 measures it for neurons in a dish); (2) fine-tune from the released weights; (3) recalibrate σ with one scalar on held-out pairs; (4) set the frame gate on validation frames; (5) monitor mean σ in production as one signal among others, re-checking with paired images whenever the imaging changes. Inference runs on a CPU and depends on nothing proprietary. A trustworthy label-free nuclear channel is a prerequisite for the organisers' goal of AI-driven OoC digital twins: a twin fed unflagged hallucinations is worse than none.

9. Reproduction

```
git clone https://github.com/hamidhosen42/AI4S-Open-Innovation-AI-for-Life-Science.git
cd AI4S-Open-Innovation-AI-for-Life-Science
python -m venv .venv && source .venv/bin/activate
pip install -r requirements.txt && pip install -e .          # exact versions:
requirements-lock.txt
python scripts/download_data.py                             # Zenodo, ~757 MB, CC BY
4.0
python scripts/download_weights.py                          # released checkpoint ->
weights/chipstain.pt

# single image (CPU is enough)
python scripts/inference.py --image demo/examples/example_bf_dense_t150.tif --weights
weights/chipstain.pt --out outputs/pred --device cpu

# full reproduction: 3 seeds x (baseline, pretrained, ChipStain) + ablation arms
bash scripts/run_all.sh 0 1 2 && bash scripts/run_ablations.sh 0 1 2
python scripts/evaluate.py --runs
runs/{baseline_unet,pretrained_l1,chipstain_nll}_s{0,1,2} --cache outputs/cache --out
outputs/multiseed
python scripts/evaluate.py --runs
runs/{baseline_unet,pretrained_l1,chipstain_nll}_s{0,1,2} --tta --cache outputs/cache --
out outputs/multiseed_tta
python scripts/evaluate.py --runs
runs/{baseline_unet,pretrained_l1,chipstain_nll}_s{0,1,2} --split val --tta --cache
outputs/cache_val --out outputs/multiseed_val
python scripts/evaluate.py --runs runs/ablate*_s{0,1,2} --out outputs/ablate
python scripts/evaluate.py --runs runs/ablate_{nll_beta0,nll_beta1}_s{0,1,2} --tta --out
outputs/ablate_tta
python scripts/multiseed_summary.py && python scripts/ensemble_eval.py && python
scripts/proliferation.py
python scripts/nucleus_uncertainty.py && python scripts/calibration.py && python
scripts/shift_test.py && python scripts/input_qc.py
curl -L -o data/raw/hela_timelapse/R05-C03-F0.tif
"https://zenodo.org/records/6139958/files/20210904_TL2%20-%20R05-C03-F0.tif?download=1"
python scripts/timelapse.py && python scripts/download_isl_neurons.py && python
scripts/neural_transfer.py
python scripts/make_figures.py && python scripts/make_figures_extra.py && python
scripts/image_register.py
python scripts/build_report.py                               # HTML always; PDF if
Chrome/Chromium is found (CHROME=/path)
python demo/app.py --weights weights/chipstain.pt            # http://localhost:7860
```

The released checkpoint reproduces its seed-0 numbers exactly on the same hardware. Retraining behaves like a new seed: expect results within the seed spread of Table 1, not identical numbers. All result files used in this report are in `report/results/`.

10. Sources, licences and AI-tool disclosure

Component	Source	Licence	Role
HeLa "Kyoto" labelled set	R. Guet, EPFL BIOP — Zenodo 10.5281/zenodo.6140064	CC BY 4.0	train / validation / test; StarDist labels as evaluation reference only
HeLa "Kyoto" time-lapse	R. Guet, EPFL BIOP — Zenodo 10.5281/zenodo.6139958	CC BY 4.0	60 h read-out of one test field
In-silico-labeling neurons (Cond. A)	Christiansen et al. [2], gs://in-silico-labeling	CC BY 4.0	neural transfer test
Cellpose bright-field nuclei models	R. Guet — Zenodo 10.5281/zenodo.6140111; Cellpose [21]	CC BY 4.0 / BSD-3	direct-segmentation competitor
ResNet-34 ImageNet-1k weights	torchvision, re-hosted as smp-hub/resnet34.imagenet on the Hugging Face Hub	code BSD-3-Clause; weights pretrained on ImageNet, whose terms may restrict commercial use	encoder initialisation
Libraries	PyTorch, segmentation_models_pytorch, timm, NumPy, SciPy, scikit-image, pandas, tiffle, Matplotlib, PyYAML, tqdm, Pillow, Gradio	BSD-3 / MIT / Apache-2.0 / PSF / MPL-2.0 / HPND	framework, analysis, demo
β-NLL, TTA, AUSE	Seitzer et al. [8]; [17]; Ilg et al. [18]	methods (re-implemented)	training objective, uncertainty, evaluation
Claude Opus 5 and Claude Opus 5.5 (Anthropic), via Claude Code	AI assistant	—	Primary implementation assistant: analysed the rules and proposed options for the direction and dataset, wrote most of the code (package, scripts, demo, notebook), ran training and evaluation on the team's laptop under the team's direction, produced the figures, drafted this report, the Writeup and the video script, and ran an automated audit of the submission. The team chose the problem, category and method, directed the work, reviewed code and outputs and takes full responsibility for the content.

No language model, foundation model, external API or paid service is used *by the method*. Released code and checkpoint: MIT; the checkpoint was trained on CC BY 4.0 data, so reuse must attribute Guet (2022), and its encoder was initialised from ImageNet-pretrained weights, so commercial users should check the ImageNet terms. This is new work created for the competition, not based on prior work by the team; it builds on the published methods cited.

References

- [1] Laissue, P. P., Alghamdi, R. A., Tomancak, P., Reynaud, E. G. & Shroff, H. Assessing phototoxicity in live fluorescence imaging. *Nature Methods* 14, 657–661 (2017).
- [2] Christiansen, E. M. et al. In silico labeling: predicting fluorescent labels in unlabeled images. *Cell* 173, 792–803 (2018).
- [3] Ounkomol, C., Seshamani, S., Malekar, M. M., Collman, F. & Johnson, G. R. Label-free prediction of three-dimensional fluorescence images from transmitted-light microscopy. *Nature Methods* 15, 917–920 (2018).
- [4] Kendall, A. & Gal, Y. What uncertainties do we need in Bayesian deep learning for computer vision? *NeurIPS* (2017).
- [5] Guet, R. Automatic labelling of HeLa "Kyoto" cells using Deep Learning tools. Zenodo, doi:10.5281/zenodo.6140064 (2022).
- [6] Guet, R. HeLa "Kyoto" cells under the scope. Zenodo, doi:10.5281/zenodo.6139958 (2022).
- [7] Ronneberger, O., Fischer, P. & Brox, T. U-Net: convolutional networks for biomedical image segmentation. *MICCAI* (2015).
- [8] Seitzer, M., Tavakoli, A., Antic, D. & Martius, G. On the pitfalls of heteroscedastic uncertainty estimation with probabilistic neural networks. *ICLR* (2022).
- [9] Iakubovskii, P. Segmentation Models PyTorch. github.com/qubvel-org/segmentation_models.pytorch (2019).
- [10] Schmidt, U., Weigert, M., Broaddus, C. & Myers, G. Cell detection with star-convex polygons. *MICCAI* (2018).
- [11] Purschke, M., Rubio, N., Held, K. D. & Redmond, R. W. Phototoxicity of Hoechst 33342 in time-lapse fluorescence microscopy. *Photochemical & Photobiological Sciences* 9, 1634–1639 (2010).
- [12] Sen, O., Saurin, A. T. & Higgins, J. M. G. The live cell DNA stain SiR-Hoechst induces DNA damage responses and impairs cell cycle progression. *Scientific Reports* 8, 7898 (2018).
- [13] Rivenon, Y. et al. Virtual histological staining of unlabelled tissue-autofluorescence images via deep learning. *Nature Biomedical Engineering* 3, 466–477 (2019).

- [14] Weigert, M. et al. Content-aware image restoration: pushing the limits of fluorescence microscopy. *Nature Methods* 15, 1090–1097 (2018).
- [15] Rotem, O., Kliper-Gross, O. & Zaritsky, A. DM4ISL: converting brightfield images into fluorescence images of organelles using DDPM with averaging mechanism and hallucination detection. *bioRxiv* preprint (2023); code at github.com/zaritskiylab/DM4ISL.
- [16] Lakshminarayanan, B., Pritzel, A. & Blundell, C. Simple and scalable predictive uncertainty estimation using deep ensembles. *NeurIPS* (2017).
- [17] Wang, G. et al. Aleatoric uncertainty estimation with test-time augmentation for medical image segmentation with convolutional neural networks. *Neurocomputing* 338, 34–45 (2019).
- [18] Ilg, E. et al. Uncertainty estimates and multi-hypotheses networks for optical flow. *ECCV* (2018).
- [19] He, K., Zhang, X., Ren, S. & Sun, J. Deep residual learning for image recognition. *CVPR* (2016).
- [20] Deng, J. et al. ImageNet: a large-scale hierarchical image database. *CVPR* (2009).
- [21] Stringer, C., Wang, T., Michaelos, M. & Pachitariu, M. Cellpose: a generalist algorithm for cellular segmentation. *Nature Methods* 18, 100–106 (2021).

Appendix A — Results by time-point (cell density)

Cell density grows over the time-lapse (frames 1 → 150 ≈ 0 → 37 h of the 60 h recording). ChipStain + TTA, mean over 3 seeds: fidelity and downstream F1 fall with density and mean σ rises with it; the real-stain segmentation falls too, so part of the drop is the segmenter.

time-point (h)	Pearson r	SSIM	seg-F1 (pred)	seg-F1 (real)	mean σ	MAE
1 (0 h)	0.806	0.888	0.813	0.829	0.0333	0.0214
10 (2.25 h)	0.799	0.874	0.791	0.820	0.0356	0.0224
50 (12.25 h)	0.780	0.823	0.740	0.767	0.0418	0.0297
100 (24.75 h)	0.745	0.769	0.629	0.720	0.0548	0.0386
150 (37.25 h)	0.710	0.714	0.568	0.689	0.0719	0.0496

Appendix B — Training curves

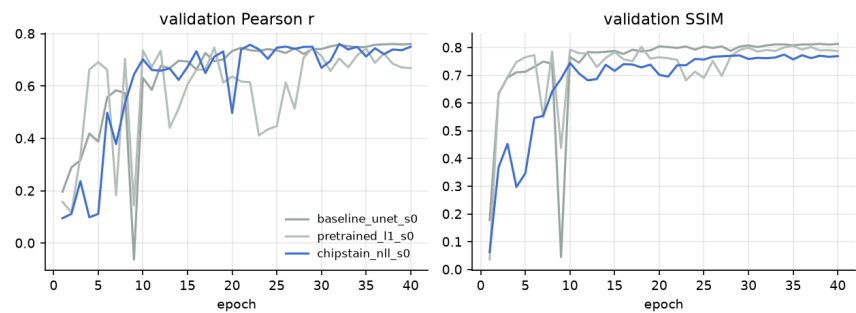


Figure B1. Validation Pearson r and SSIM per epoch for the three main configurations (seed 0). The validation curves oscillate (e.g. the pretrained-L1 run drops sharply at some epochs); the best-validation checkpoint hides part of this instability, which shows instead as seed-to-seed spread in §6.1.

Appendix C — Additional qualitative comparisons

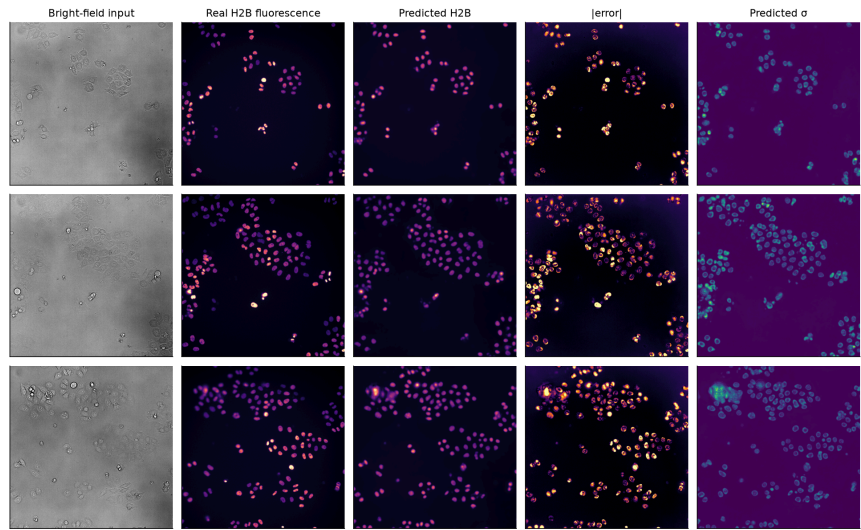


Figure C1. The test images of Figure 3 predicted by the plain U-Net with TTA; the last column is its TTA disagreement, used as its uncertainty.

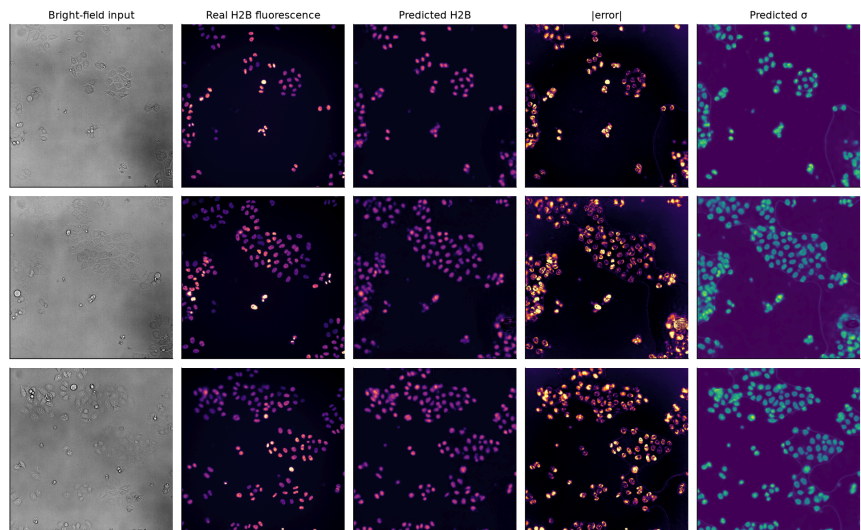


Figure C2. ChipStain without TTA (single forward pass): the learned σ alone.

Appendix D — Requirement checklist

Challenge requirement	Where
Category declaration	First line of the Kaggle Writeup and of this report
Demo video (≤ 5 min, public)	Kaggle Writeup
Public repository: code, README, environment, weights, entry scripts, I/O description, reproduction	GitHub repository; README; requirements.txt / requirements-lock.txt / pyproject.toml; scripts/download_weights.py; scripts/inference.py; §9
Data sources, licences, processing, compliance	§3, DATA.md, Appendix E
Methods, implementation, experiments, results	§4–§6
Reliability and limitations; impact	§7; §8
External models, libraries, datasets, AI tools	§10
Team composition	Header of this report (team Hack2Publish, three members, Computer Science and Engineering)

Appendix E — Image register

Every image used in this report, in the Kaggle Writeup (cover, thumbnail, gallery) and in the repository, with its pixel source and the changes made. All microscopy pixels come from CC BY 4.0 datasets; figures containing them are CC BY 4.0 derivatives with the attribution given. Charts contain no dataset pixels. (Also in report/IMAGES.md.)

file	pixels	shows	pixel source	changes
demo/examples/example_bf_dense_t150.tif	540×540	Demo inputs and matching real H2B	HeLa "Kyoto" test well R05-C03 — R. Guiet (EPFL BIOP), Zenodo 10.5281/zenodo.6140064, CC BY 4.0	unmodified (H2B: channel 1 of the original 2-channel file)
demo/examples/example_bf_dense_t150_blur1px.tif	540×540	Demo input with simulated defocus	HeLa "Kyoto" test well R05-C03 — R. Guiet (EPFL BIOP), Zenodo 10.5281/zenodo.6140064, CC BY 4.0	Gaussian blur $\sigma = 1$ px applied by this project
demo/examples/example_bf_sparse_t010.tif	540×540	Demo inputs and matching real H2B	HeLa "Kyoto" test well R05-C03 — R. Guiet (EPFL BIOP), Zenodo 10.5281/zenodo.6140064, CC BY 4.0	unmodified (H2B: channel 1 of the original 2-channel file)
demo/examples/example_h2b_real_t010.tif	540×540	Demo inputs and matching real H2B	HeLa "Kyoto" test well R05-C03 — R. Guiet (EPFL BIOP), Zenodo 10.5281/zenodo.6140064, CC BY 4.0	unmodified (H2B: channel 1 of the original 2-channel file)
demo/examples/example_h2b_real_t150.tif	540×540	Demo inputs and matching real H2B	HeLa "Kyoto" test well R05-C03 — R. Guiet (EPFL BIOP), Zenodo 10.5281/zenodo.6140064, CC BY 4.0	unmodified (H2B: channel 1 of the original 2-channel file)
report/figures/demo_panel_dense_t150.png	1560×546	Inference panel: bright-field, predicted nuclei, σ	HeLa "Kyoto" test well R05-C03 — R. Guiet (EPFL BIOP), Zenodo 10.5281/zenodo.6140064, CC BY 4.0	intensity-normalised, colour-mapped; prediction is model output
report/figures/demo_panel_dense_t150_blur1px.png	1560×546	Inference panel: bright-field, predicted nuclei, σ	HeLa "Kyoto" test well R05-C03 — R. Guiet (EPFL BIOP), Zenodo 10.5281/zenodo.6140064, CC BY 4.0	intensity-normalised, colour-mapped; prediction is model output
report/figures/demo_panel_sparse_t010.png	1560×546	Inference panel: bright-field, predicted nuclei, σ	HeLa "Kyoto" test well R05-C03 — R. Guiet (EPFL BIOP), Zenodo 10.5281/zenodo.6140064, CC BY 4.0	intensity-normalised, colour-mapped; prediction is model output
report/figures/failure_gallery.png	1644×1399	Worst test images across seeds: input, real H2B, prediction, error , σ	HeLa "Kyoto" test well R05-C03 — R. Guiet (EPFL BIOP), Zenodo 10.5281/zenodo.6140064, CC BY 4.0	full fields, intensity-normalised, colour-mapped; predictions are model output
report/figures/multiseed.png	1999×518	3-seed comparison of all configurations	None (chart of numbers computed by this project)	—
report/figures/neural_examples.png	1941×418	Neural transfer: bright-field, real Hoechst (DAPI_WIDEFIELD channel), predictions, σ , fine-tuning curve	Human iPSC-derived motor neurons (Condition A) — Christiansen et al., Cell 2018, gs://in-silico-labeling , CC BY 4.0	rescaled to the HeLa pixel scale (factor chosen on validation wells), intensity-normalised, colour-mapped
report/figures/neural_finetune.png	1942×531	Neural transfer: bright-field, real Hoechst (DAPI_WIDEFIELD channel), predictions, σ , fine-tuning curve	Human iPSC-derived motor neurons (Condition A) — Christiansen et al., Cell 2018, gs://in-silico-labeling , CC BY 4.0	rescaled to the HeLa pixel scale (factor chosen on validation wells), intensity-normalised, colour-mapped
report/figures/nucleus_gate.png	1603×531	Per-nucleus quality-gate curves	None (chart of numbers computed by this project)	—
report/figures/proliferation.png	1688×565	Growth curves / doubling times from label-free counts	None (chart of numbers computed by this project)	—
report/figures/qualitative_baseline_unet_s0_tta.png	1950×1215	Test images: bright-field input, real H2B, prediction, error , predicted σ	HeLa "Kyoto" test well R05-C03 — R. Guiet (EPFL BIOP), Zenodo 10.5281/zenodo.6140064, CC BY 4.0	full 540×540 fields, intensity-normalised, colour-mapped; predictions are model output
report/figures/qualitative_chipstain_nll_s0.png	1950×1215	Test images: bright-field input,	HeLa "Kyoto" test well R05-C03 — R. Guiet (EPFL	full 540×540 fields, intensity-normalised, colour-mapped;

		real H2B, prediction, error , predicted σ	BIOP), Zenodo 10.5281/zenodo.6140064, CC BY 4.0	predictions are model output
report/figures/qualitative_chipstain_nll_s0_tta.png	1950×1215	Test images: bright-field input, real H2B, prediction, error , predicted σ	HeLa "Kyoto" test well R05-C03 — R. Guiet (EPFL BIOP), Zenodo 10.5281/zenodo.6140064, CC BY 4.0	full 540×540 fields, intensity-normalised, colour-mapped; predictions are model output
report/figures/shift_test.png	1690×701	Accuracy and σ under simulated distribution shifts	None (chart of numbers computed by this project)	—
report/figures/timelapse.png	1347×769	60 h label-free nuclear read-out of one field	HeLa "Kyoto" time-lapse, well R05-C03 field F0 — R. Guiet (EPFL BIOP), Zenodo 10.5281/zenodo.6139958, CC BY 4.0	intensity-normalised, colour-mapped; counts are model/segmentation output
report/figures/timelapse_strip.png	1644×1042	60 h label-free nuclear read-out of one field	HeLa "Kyoto" time-lapse, well R05-C03 field F0 — R. Guiet (EPFL BIOP), Zenodo 10.5281/zenodo.6139958, CC BY 4.0	intensity-normalised, colour-mapped; counts are model/segmentation output
report/figures/training_curves.png	1275×480	Validation curves per epoch	None (chart of numbers computed by this project)	—
report/figures/uncertainty_baselines.png	1091×599	Uncertainty vs uncertainty-free proxies	None (chart of numbers computed by this project)	—
writeup/assets/cover_560x280.png	560×280	Kaggle cover / thumbnail	HeLa "Kyoto" test well R05-C03 — R. Guiet (EPFL BIOP), Zenodo 10.5281/zenodo.6140064, CC BY 4.0	380×380 crop of field F3 t150, resized, colour-mapped, title text added
writeup/assets/gallery_1_qualitative.png	1950×1215	Kaggle media-gallery copies of report figures	HeLa "Kyoto" test well R05-C03 — R. Guiet (EPFL BIOP), Zenodo 10.5281/zenodo.6140064, CC BY 4.0 (photographic panels) / none (charts)	copies of report/figures/*
writeup/assets/gallery_2_multiseed.png	1999×518	Kaggle media-gallery copies of report figures	HeLa "Kyoto" test well R05-C03 — R. Guiet (EPFL BIOP), Zenodo 10.5281/zenodo.6140064, CC BY 4.0 (photographic panels) / none (charts)	copies of report/figures/*
writeup/assets/gallery_3_shift_test.png	1690×701	Kaggle media-gallery copies of report figures	HeLa "Kyoto" test well R05-C03 — R. Guiet (EPFL BIOP), Zenodo 10.5281/zenodo.6140064, CC BY 4.0 (photographic panels) / none (charts)	copies of report/figures/*
writeup/assets/gallery_4_timelapse.png	1347×769	Kaggle media-gallery copy of the time-lapse figure	HeLa "Kyoto" time-lapse, well R05-C03 field F0 — R. Guiet (EPFL BIOP), Zenodo 10.5281/zenodo.6139958, CC BY 4.0	copy of report/figures/timelapse.png
writeup/assets/gallery_5_demo_panel.png	1560×546	Kaggle media-gallery copies of report figures	HeLa "Kyoto" test well R05-C03 — R. Guiet (EPFL BIOP), Zenodo 10.5281/zenodo.6140064, CC BY 4.0 (photographic panels) / none (charts)	copies of report/figures/*
writeup/assets/gallery_6_proliferation.png	1688×565	Kaggle media-gallery copies of report figures	HeLa "Kyoto" test well R05-C03 — R. Guiet (EPFL BIOP), Zenodo 10.5281/zenodo.6140064, CC BY 4.0 (photographic panels) / none (charts)	copies of report/figures/*
writeup/assets/gallery_7_neural.png	1941×418	Kaggle media-gallery copy of the neural-transfer figure	Human iPSC-derived motor neurons (Condition A) — Christiansen et al., Cell 2018, gs://in-silico-labeling , CC BY 4.0	copy of report/figures/neural_examples.png